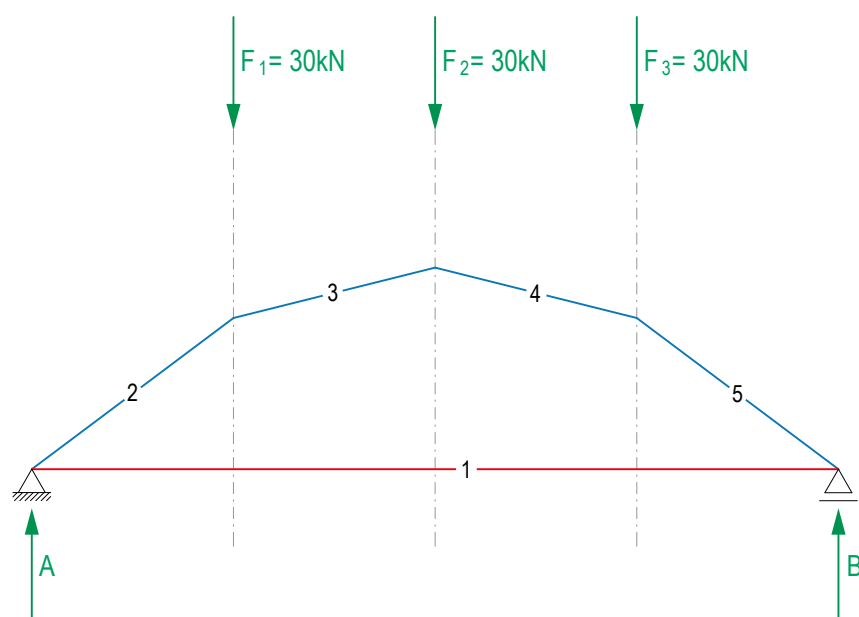
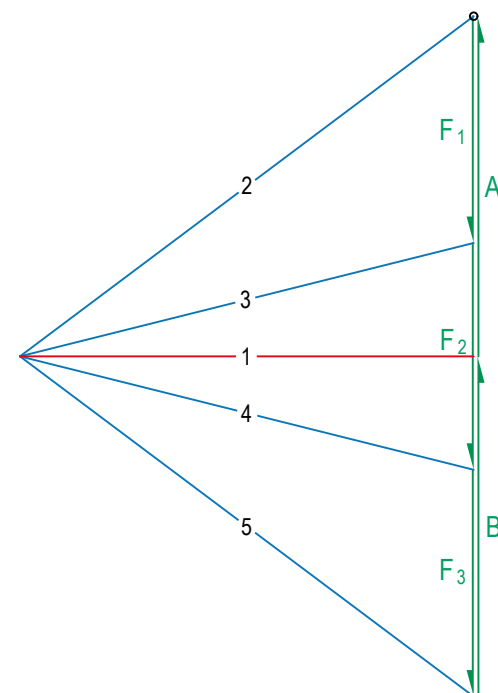


Additional Form-Finding: Arch-Cable

Task 1.1 Find the arch-cable-structure that forms under the given loading situation. The tension force in the bottom chord equals 60kN. Draw the corresponding force diagram. Use red for tension and blue for compression.



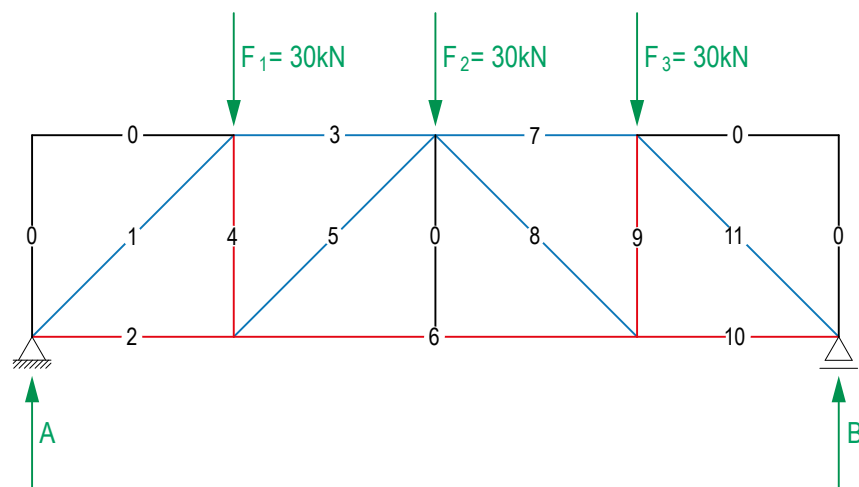
form diagram 1:100



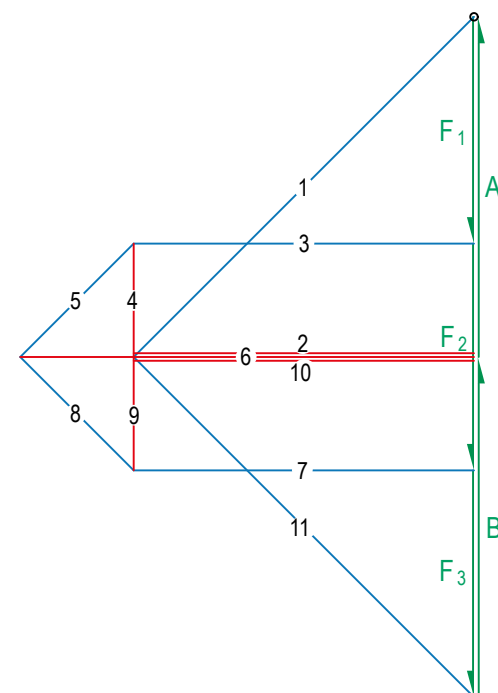
force diagram 1cm $\hat{=}$ 10kN

Additional Analysis: Truss 1

Task 1.2 This loading case corresponds to that of 1.1, but here a truss is analysed. Draw the corresponding force diagram for the given situation. First identify possible zero members. Indicate tension forces with red and compression forces with blue.



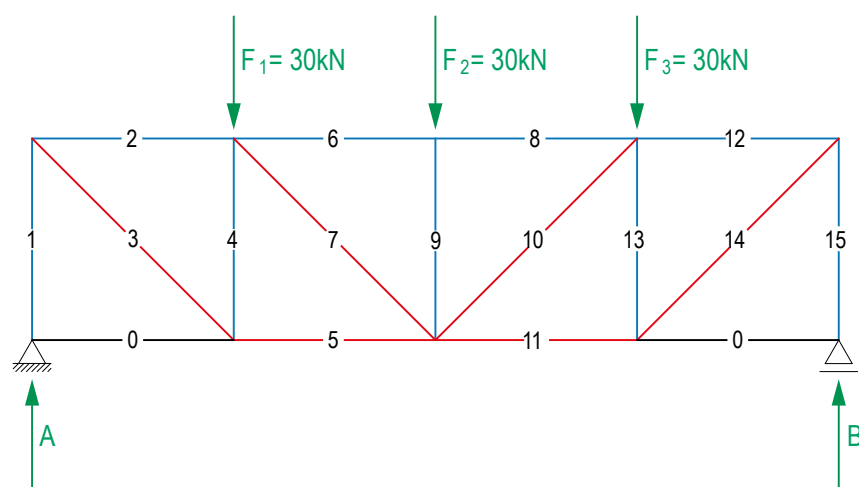
form diagram 1:100



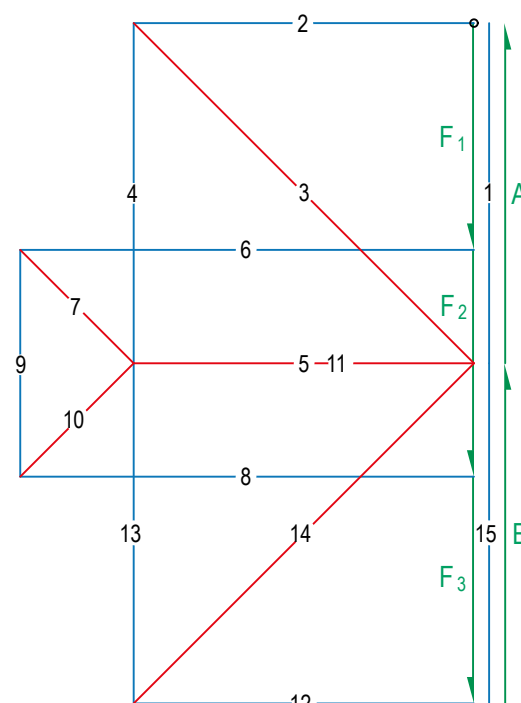
force diagram 1cm $\hat{=}$ 10kN

Additional Analysis: Truss 2

Task 1.3 This loading case corresponds to that of 1.1 and 1.2, but here a different truss is analysed. Draw the corresponding force diagram for the given situation. Indicate tension forces with red and compression forces with blue.



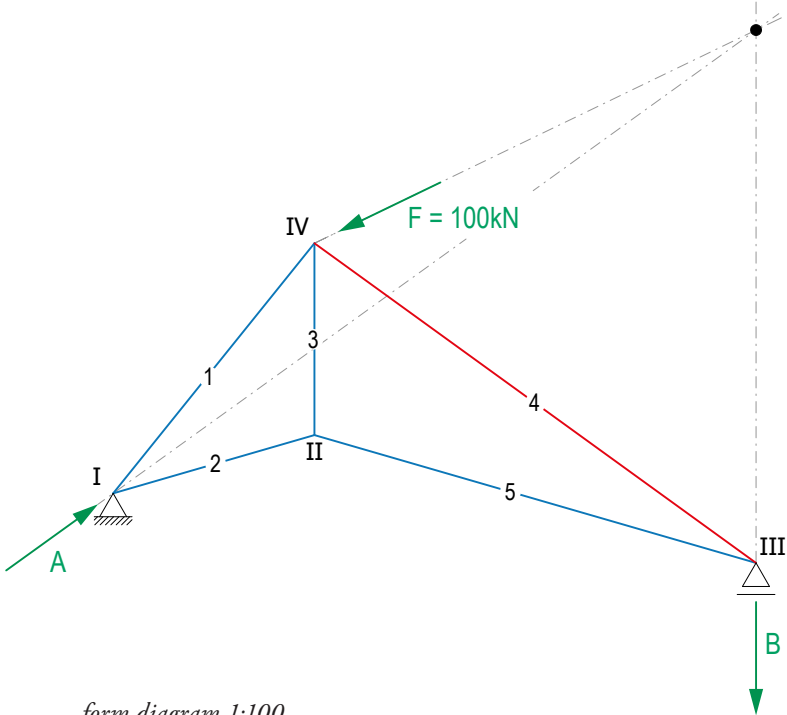
form diagram 1:100



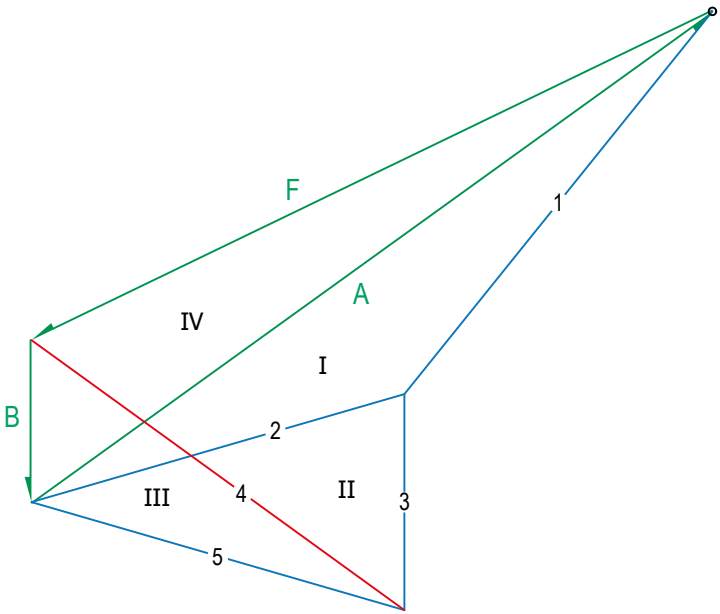
force diagram 1cm $\hat{=}$ 10kN

Additional Truss

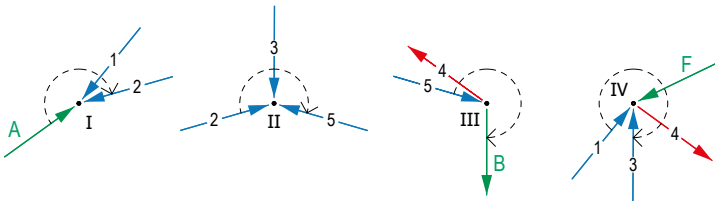
Task 3 Draw the corresponding force diagram for the given truss. First, find the global equilibrium. Indicate tension forces with red and compression forces with blue.



form diagram 1:100



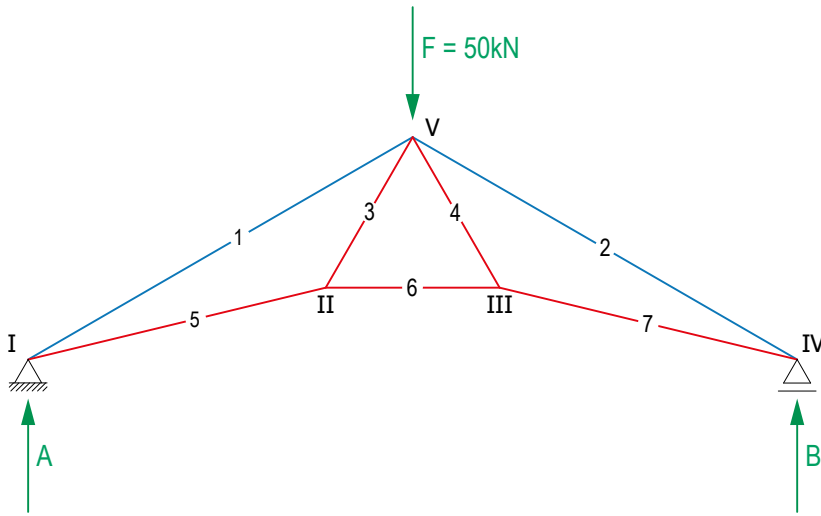
force diagram 1cm $\triangleq$ 10kN



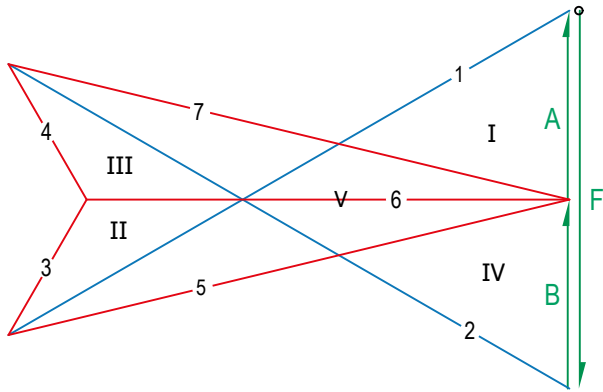
subsystems

Additional Truss

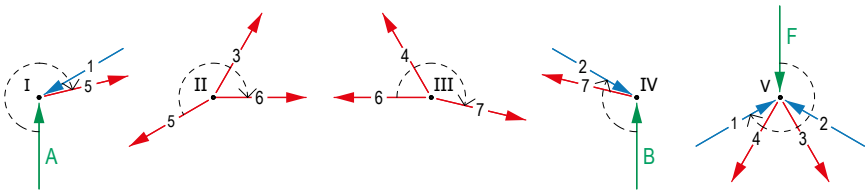
Task 4 Draw the corresponding force diagram for the given truss. Indicate tension forces with red and compression forces with blue.



form diagram 1:100



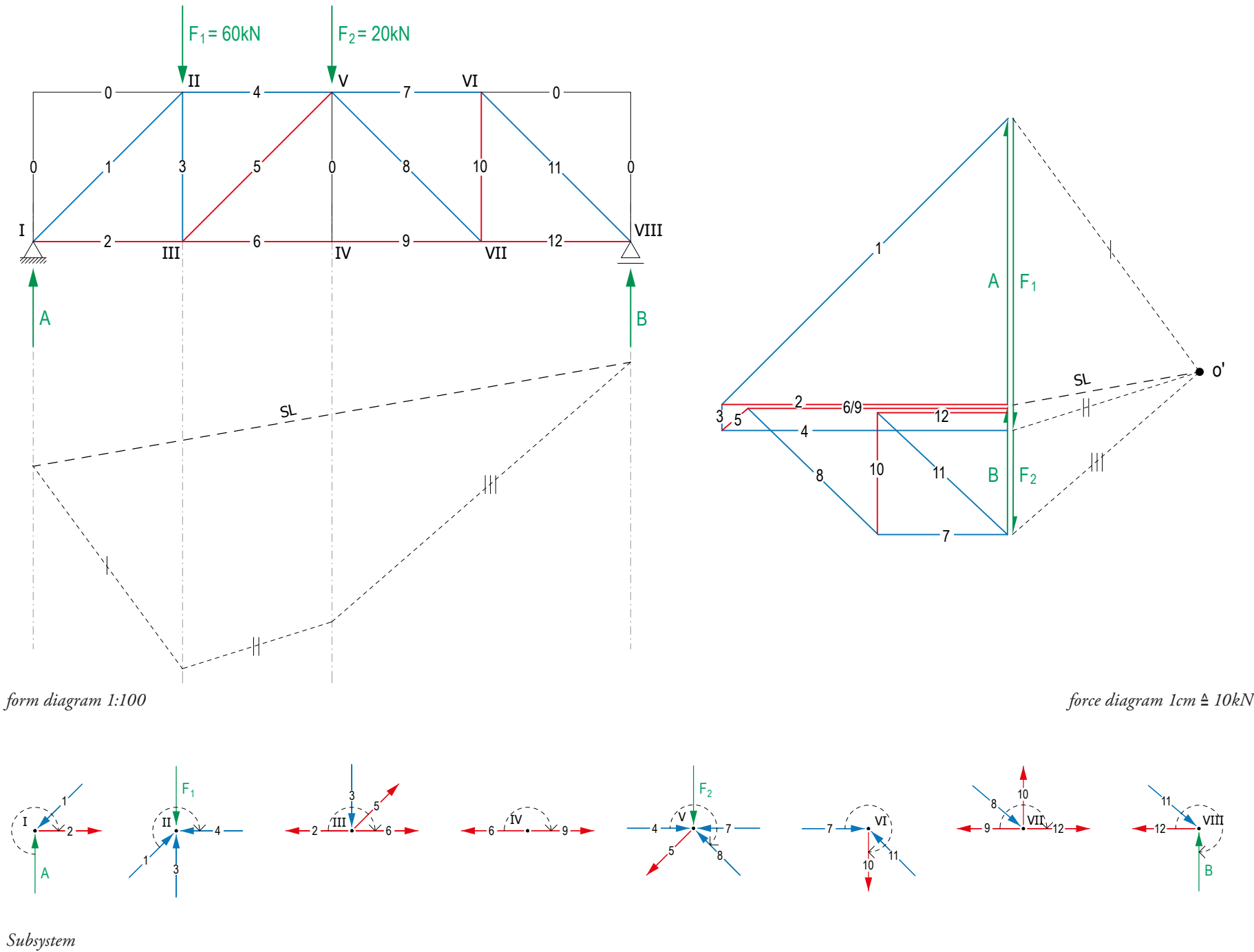
force diagram 1cm $\triangleq$ 10kN



subsystem

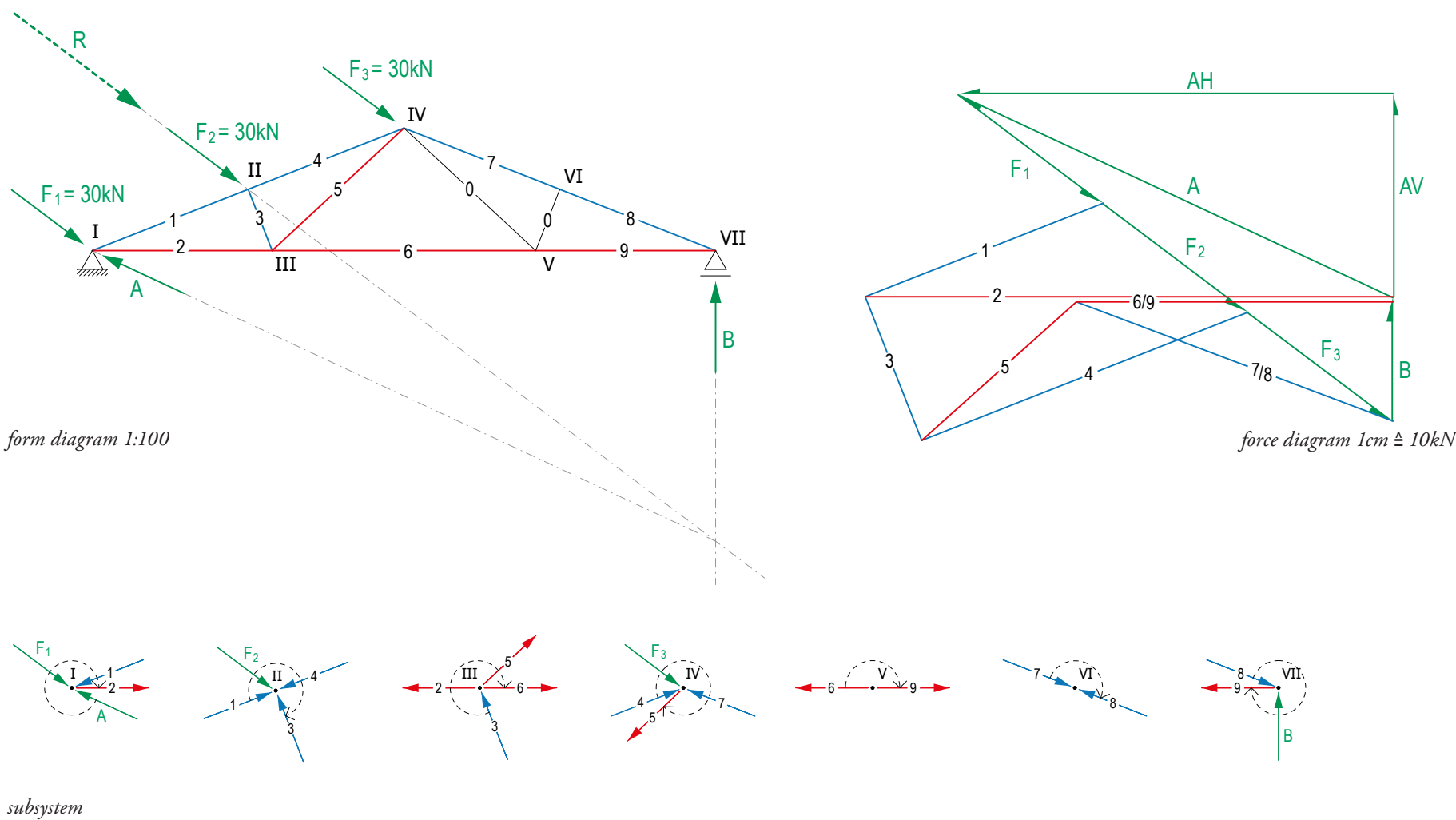
Additional Truss under non-Uniformly Distributed Load

Task 1 Determine both reaction forces A and B for the following truss. Draw the corresponding force diagram for the given case. Indicate tension forces with red and compression forces with blue.



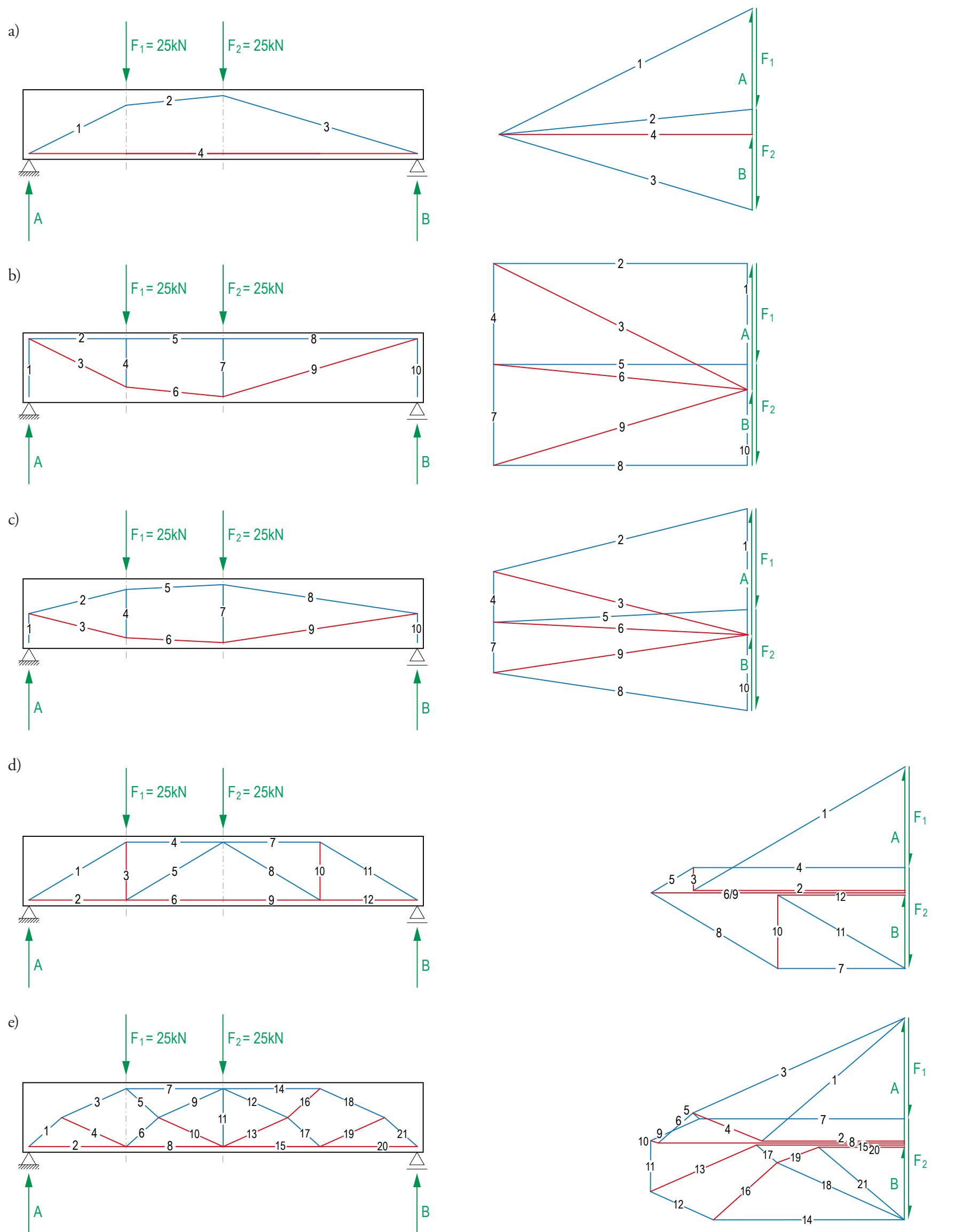
Additional Truss under Inclined Load

Task 2 Determine the resultant force R and both reaction forces A and B for the following truss. Draw the corresponding force diagram for the given case. Indicate tension forces with red and compression forces with blue.



Task 1 Internal Force Distribution in a Simple Beam

Draw the force diagrams for the given distribution of internal forces for the beams in a) - c). For the cases d) and e), the force diagram is already provided. Indicate tension forces with red and compression forces with blue. Compare in f) the distribution of internal forces in a) - c) with the versions in d) and e).



form diagrams 1:100

force diagrams 1cm $\hat{=}$ 10kN

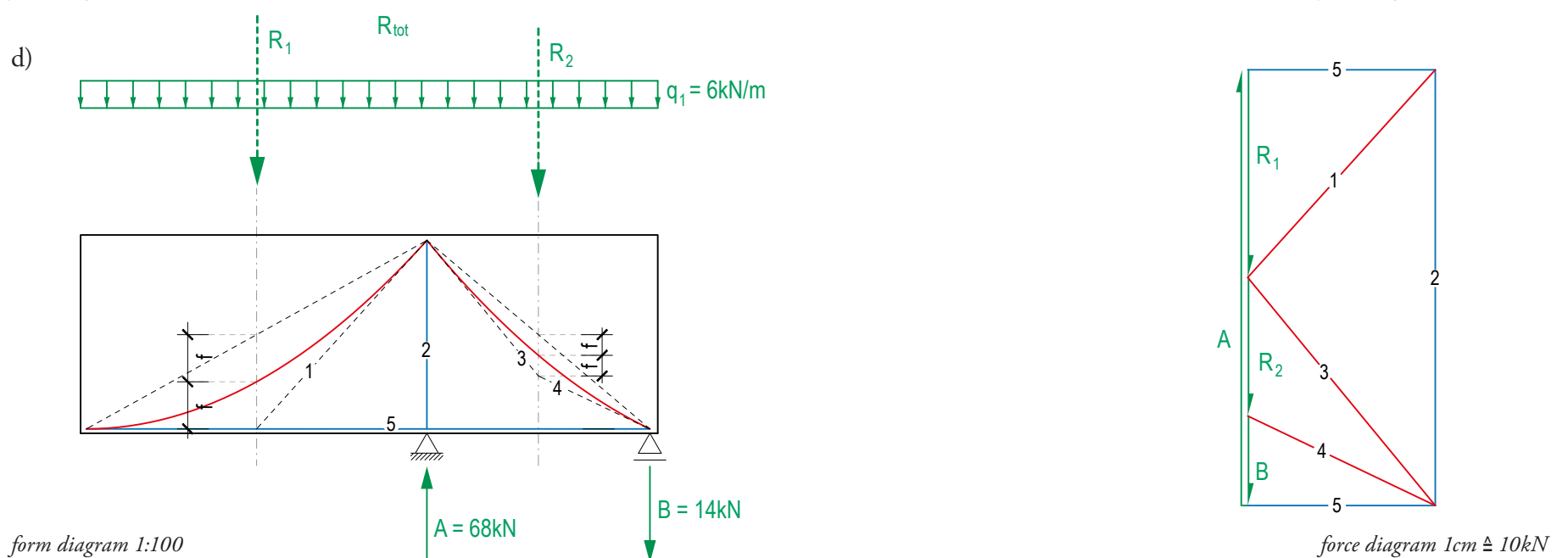
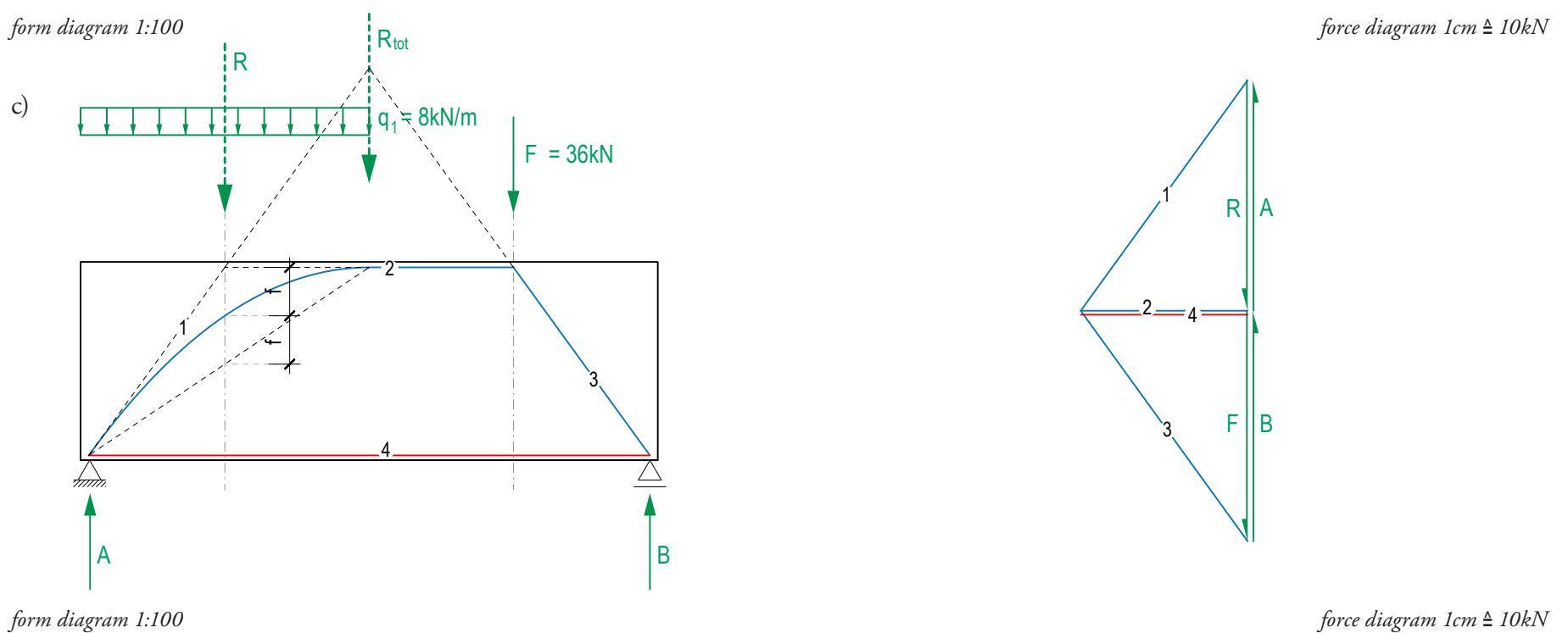
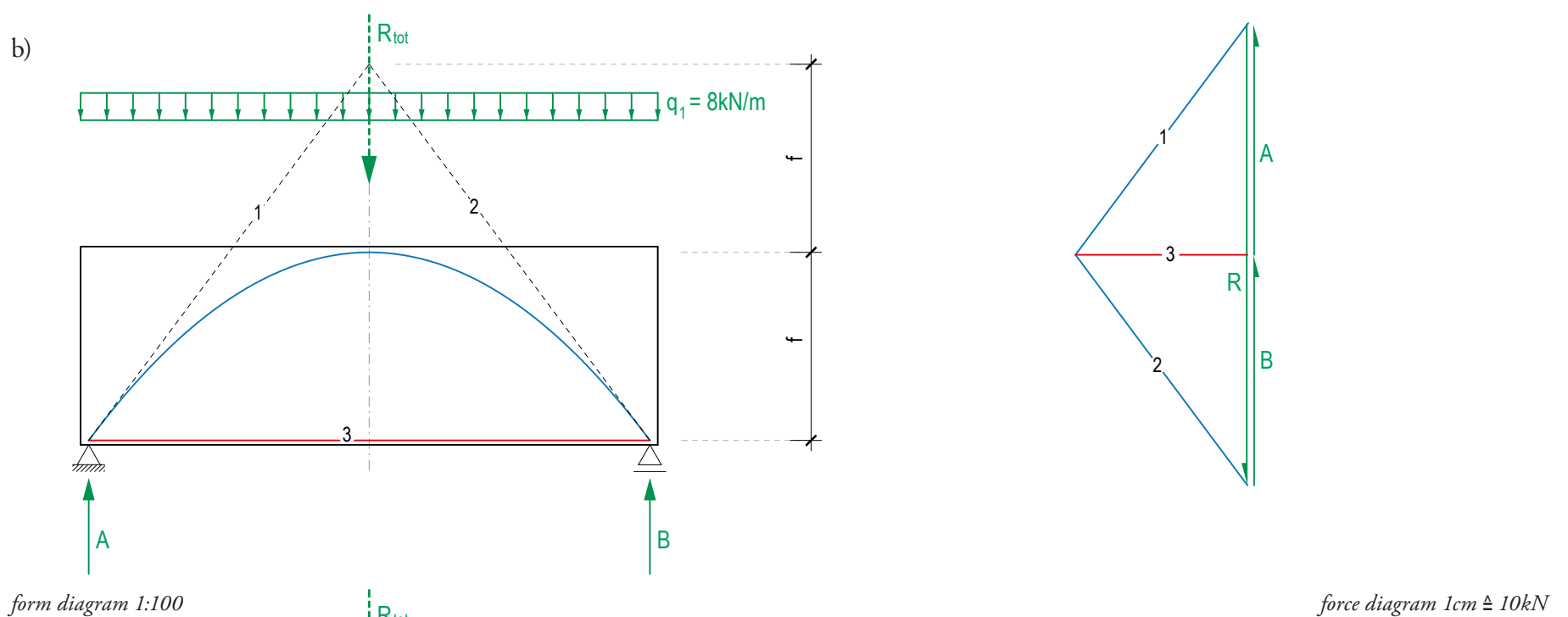
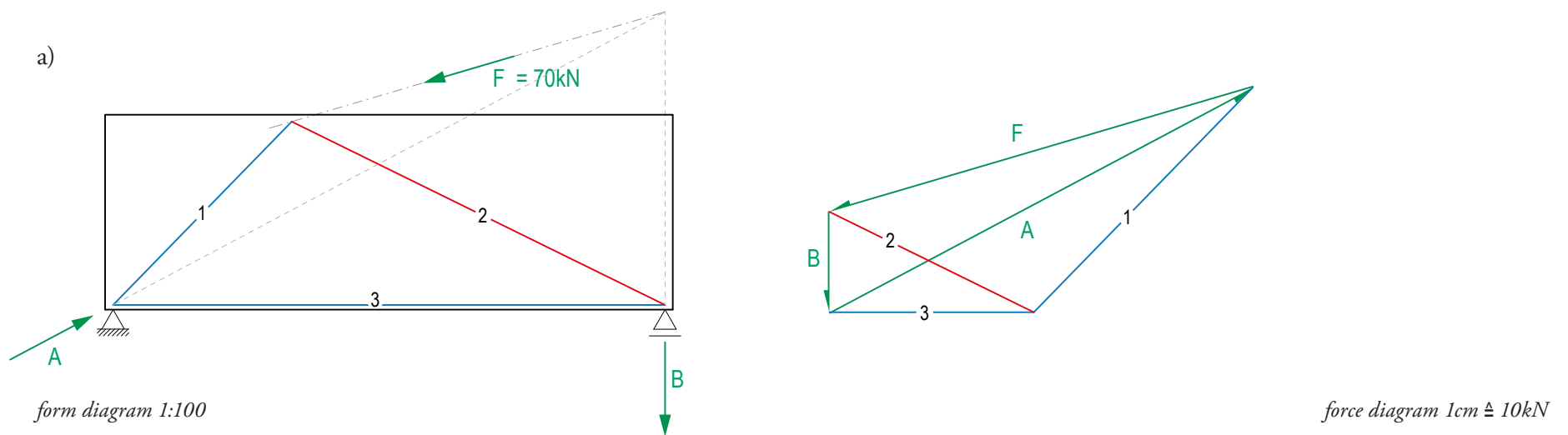
f)

The cases a) to c) show solutions with cable-arch-constructions within the beams. For such a solution within reinforced concrete, the tension elements should be pre-stressed. Otherwise large cracks can occur in the concrete. In cases d) and e) there is a truss-like structure formed inside the beam. The forces are distributed better, as well as possible cracks (a lot of small cracks instead of big ones). This way pre-stress can be avoided.

Additional Statically determined supported wall slab

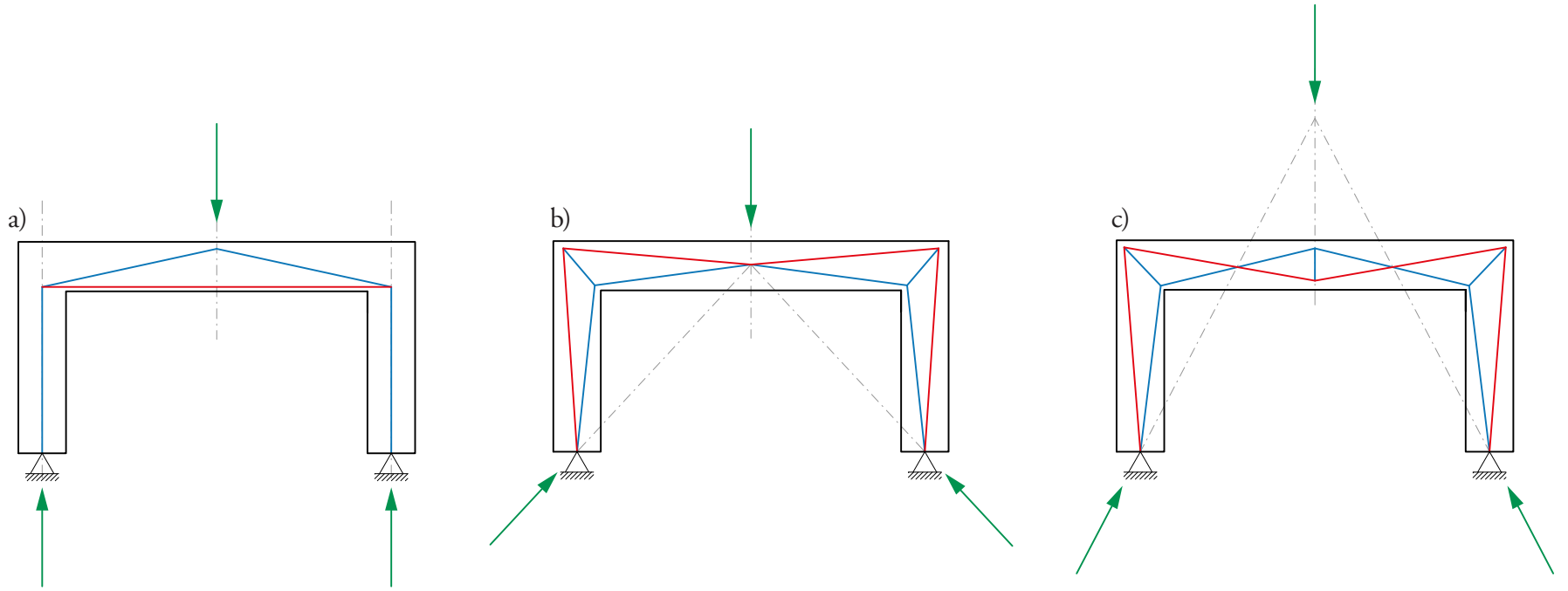
Task 1 Four equal wall slabs in reinforced concrete but with different supports or loads are given. In situations a) to c) draw a possible internal force distribution as an arch-cable-structure with the aid of the force diagram.

Draw the corresponding force diagram to the given force distribution in d). Indicate tension forces with red and compression forces with blue.



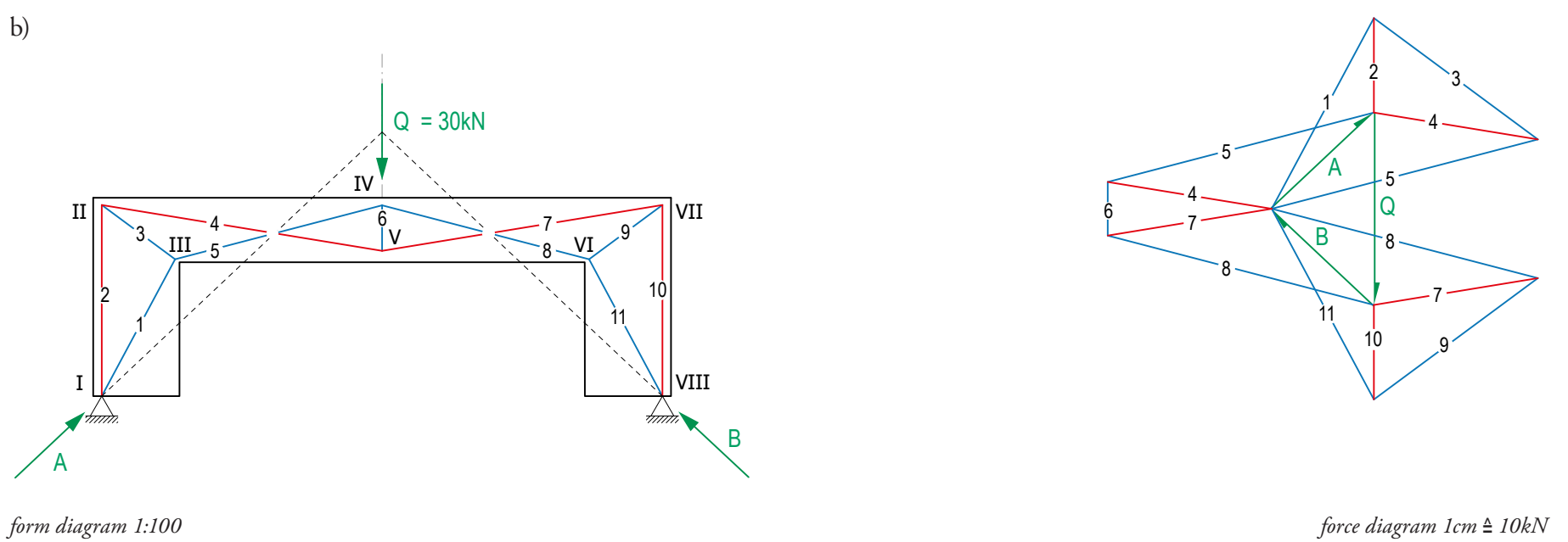
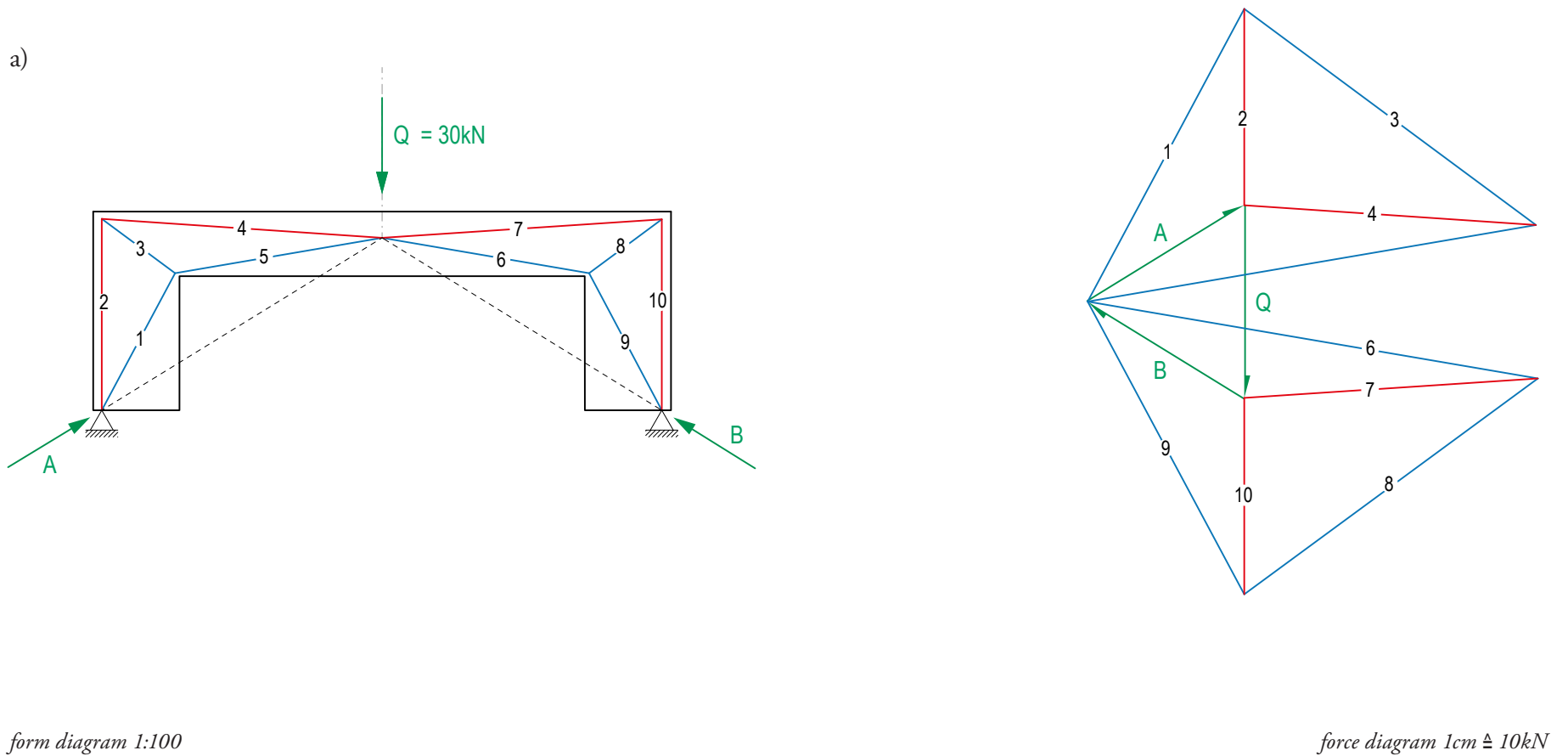
Task 3 Qualitative Inner Force Distribution in a Reinforced Concrete Frame

Three identical reinforced concrete frames with different support conditions are given. Draw the possible inner force distribution for each case. Indicate tension forces with red and compression forces with blue.



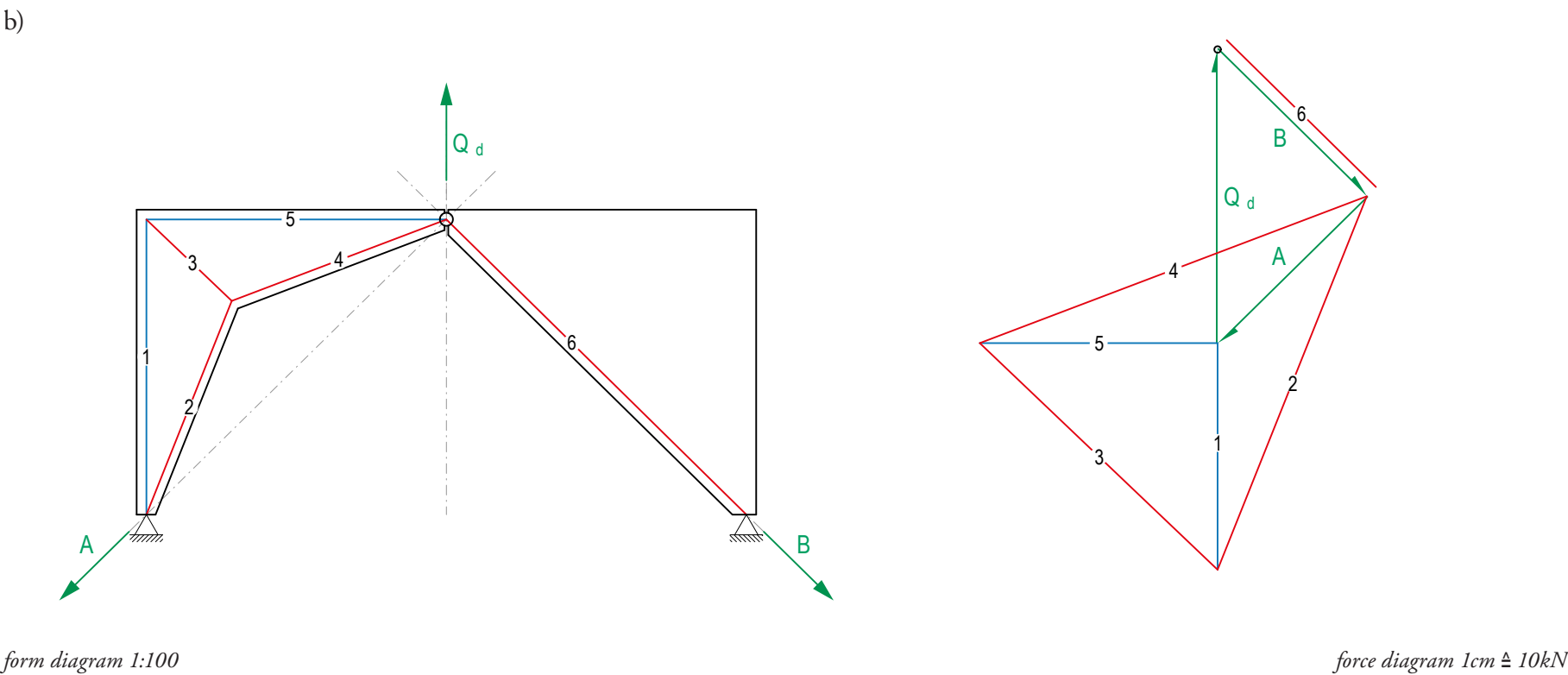
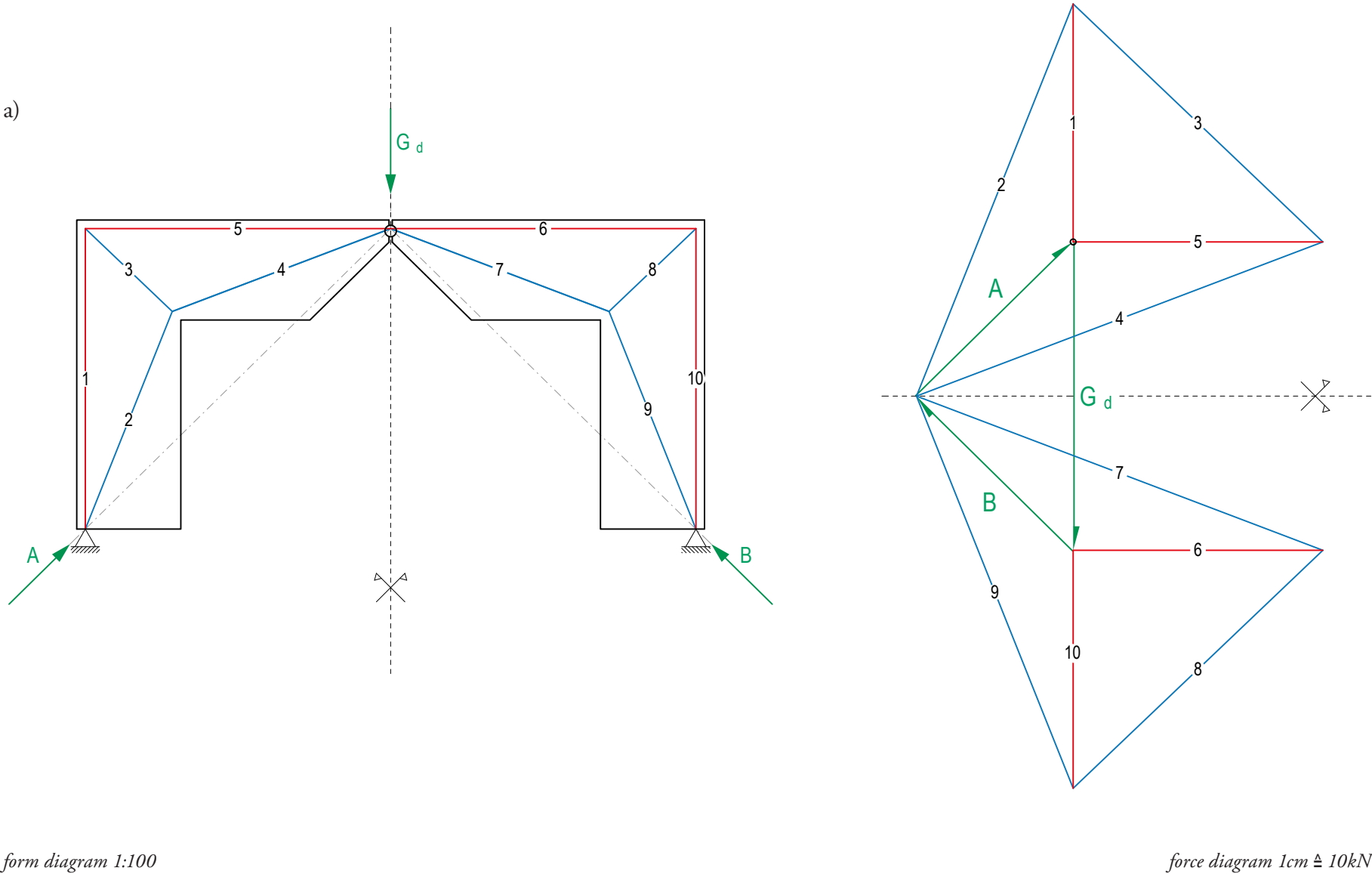
Additional Statically indetermined supported frame

Task 1 Two equal frames in reinforced concrete with different supports are given. In situation a) draw a possible internal force distribution as an arch-cable-construction with the aid of the force diagram. Draw the corresponding force diagram to the given force distribution in b). Indicate in both a) and b) tension forces with red and compression forces with blue.



Task 1 Internal Force Flow in Frames

Draw a possible internal force flow for the reinforced concrete frame with help of the thrust line. Draw the corresponding force diagram and indicate tension forces with red, compression forces with blue and reaction forces with green.



Additional Axial Force Proof

Task 2 Check whether the reinforcement in the reinforced concrete in additional task 1a) can withstand the relevant tension force. The reinforcement has a round section with a diameter of 16 mm and is made of steel S235.

N_d

=

44 kN

f_{td}

=

$f_{tk} / \gamma_M = 235 \text{ N/mm}^2 / 1.05 = 223.8 \text{ N/mm}^2$

A_{eff}

=

$(D/2)^2 \cdot \pi = 64 \text{ mm}^2 \cdot \pi = 201.1 \text{ mm}^2$

N_{allow}

=

$f_{td} \cdot A_{eff} = 223.8 \text{ N/mm}^2 \cdot 201.1 \text{ mm}^2 = 44'998 \text{ N}$

N_d

=

44 kN

≤

45 kN

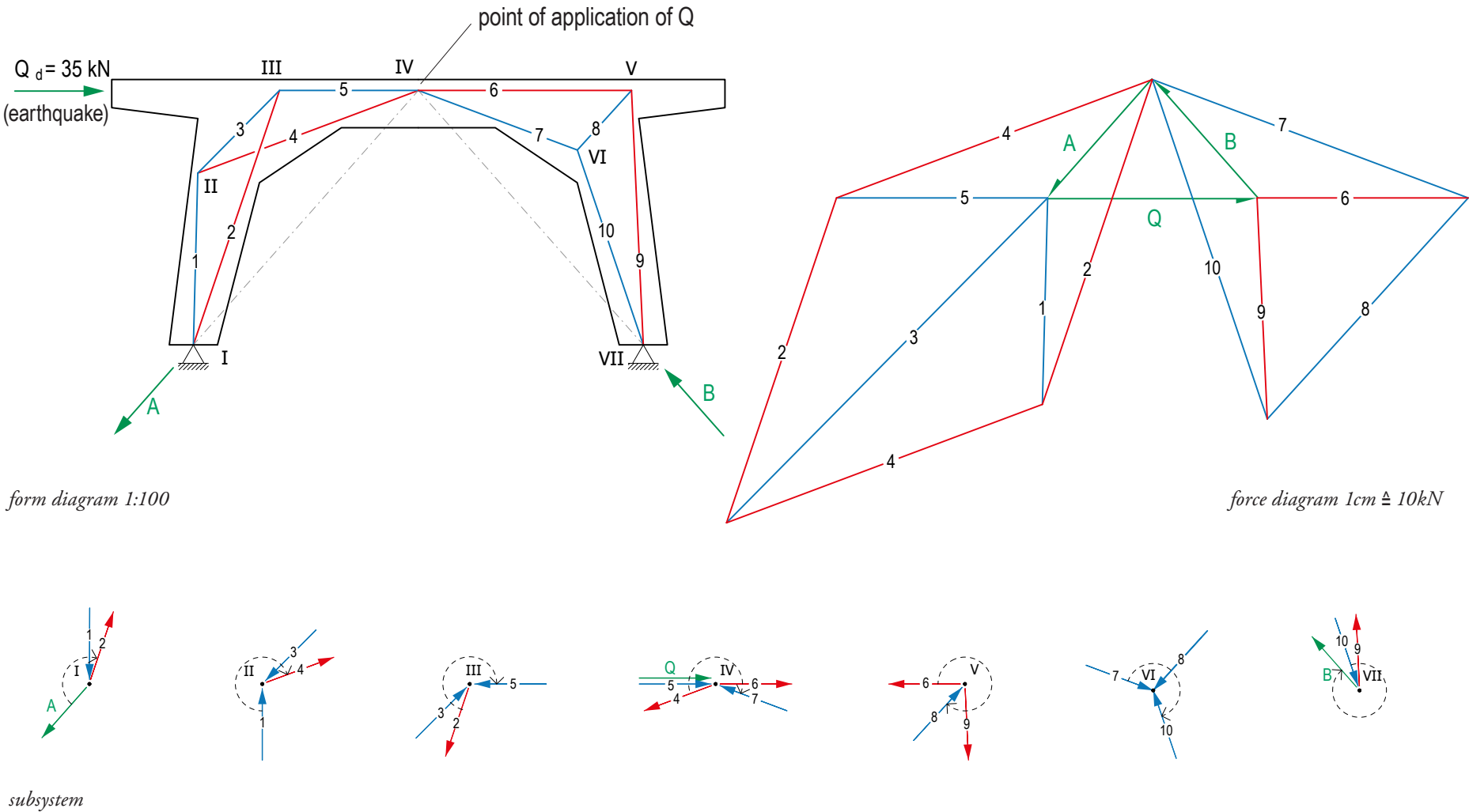
=

N_{allow}

Das Profil ist ausreichend gross dimensioniert für die massgebende Zugkraft.
The profile is sufficiently dimensioned for the relevant tension force.

Additional Reinforced Concrete Frame

Task 3 Given is the possible distribution of the internal forces for a reinforced concrete frame. Draw the corresponding force diagram. Indicate tension forces with red and compression forces with blue.



Additional Dimensioning

- Task 2
- a) Dimension the reinforcement within the reinforced concrete for the relevant tension force of task 2 a). Use steel S235 to calculate the diameter and round the result off to mm. (Round up!)
 - b) Verify whether the frame in task 2 b) can keep up with the relevant compression force. The slab is 8 cm thick and is constructed in concrete C12/15. Presume the force would affect the frame over the width of 10 cm.

N_d

=

57.4

kN

f_{td}

=

$f_{tk} / \gamma_M = 235 \text{ N/mm}^2 / 1.05 = 223.8$

N/mm²

A

=

$N_d / f_{td} = 57400 \text{ N} / 223.8 \text{ N/mm}^2 = 256.48$

mm²

D

=

$\sqrt{4 \cdot A / \pi} = \sqrt{4 \cdot 256.48 \text{ mm}^2 / \pi} = 18.07$

mm

$\approx 19 \text{ mm}$

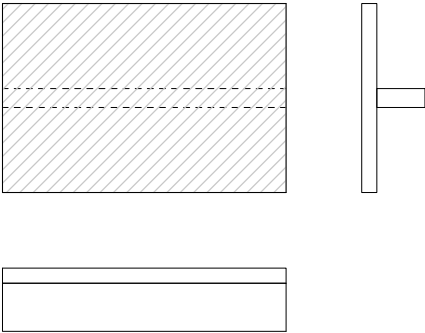
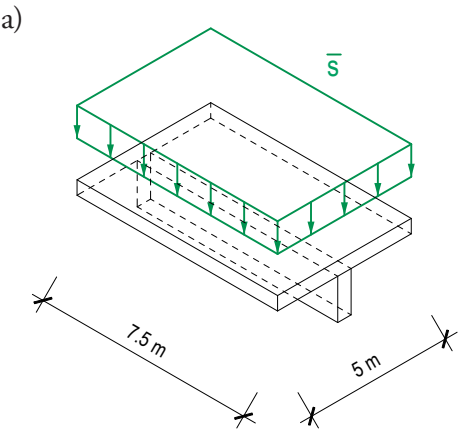
Task 1 Load-influenced Area

The plate is loaded by a dead area load of $\bar{s}_k = 1 \text{ kN/m}^2$. Calculate the design value of the constant area load.

Draw the relevant load-influenced area into the floor plan for both of the following cases, a) and b).

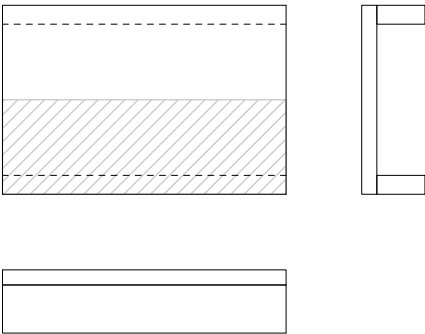
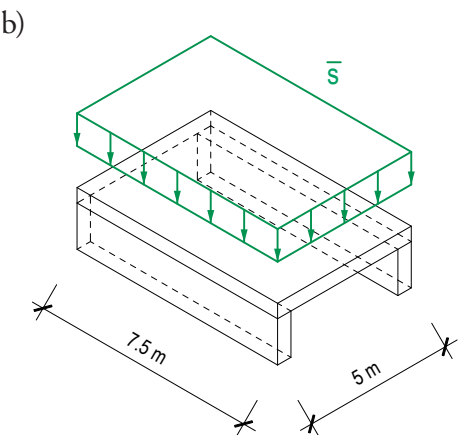
Then calculate the point load R and the line load g over the relevant beam and complete the table.

$$\bar{s}_d = \bar{s}_k \cdot \gamma_G$$
$$\bar{s}_d = 1 \text{ kN/m}^2 \cdot 1.35 = 1.35 \text{ kN/m}^2$$



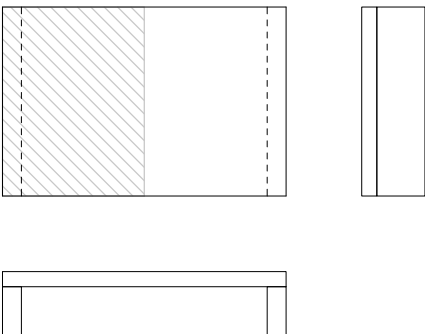
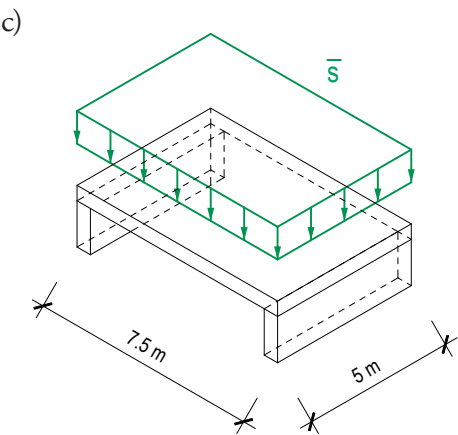
A	37.5 m²
$\bar{s}_d$	1.35 kN/m²
R	50.63 kN
g	6.75 kN/m

$$A = 7.5\text{m} \cdot 5\text{m} = 37.5 \text{ m}^2$$
$$R = \bar{s}_d \cdot A = 1.35 \text{ kN/m}^2 \cdot 37.5 \text{ m}^2 = 50.63 \text{ kN}$$
$$g = R / l = 50.63 \text{ kN} / 7.5 \text{ m} = 6.75 \text{ kN/m}$$



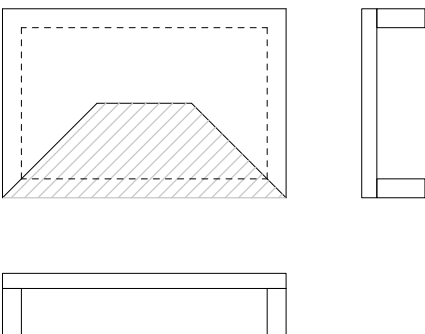
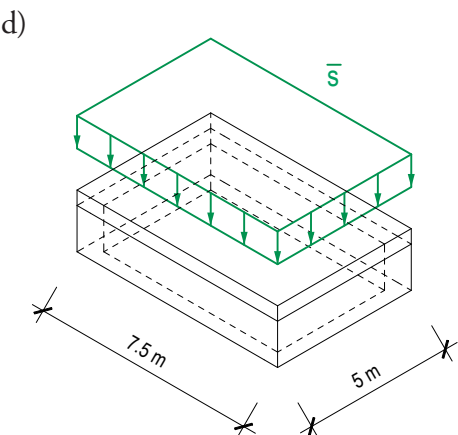
A	18.75 m²
$\bar{s}_d$	1.35 kN/m²
R	25.31 kN
g	3.38 kN/m

$$A = 2.5\text{m} \cdot 7.5\text{m} = 18.75 \text{ m}^2$$
$$R = \bar{s}_d \cdot A = 1.35 \text{ kN/m}^2 \cdot 18.75 \text{ m}^2 = 25.31 \text{ kN}$$
$$g = R / l = 25.31 \text{ kN} / 7.5 \text{ m} = 3.38 \text{ kN/m}$$



A	18.75 m²
$\bar{s}_d$	1.35 kN/m²
R	25.31 kN
g	5.06 kN/m

$$A = 3.75\text{m} \cdot 5\text{m} = 18.75 \text{ m}^2$$
$$R = \bar{s}_d \cdot A = 1.35 \text{ kN/m}^2 \cdot 18.75 \text{ m}^2 = 25.31 \text{ kN}$$
$$g = R / l = 25.31 \text{ kN} / 5\text{m} = 5.06 \text{ kN/m}$$



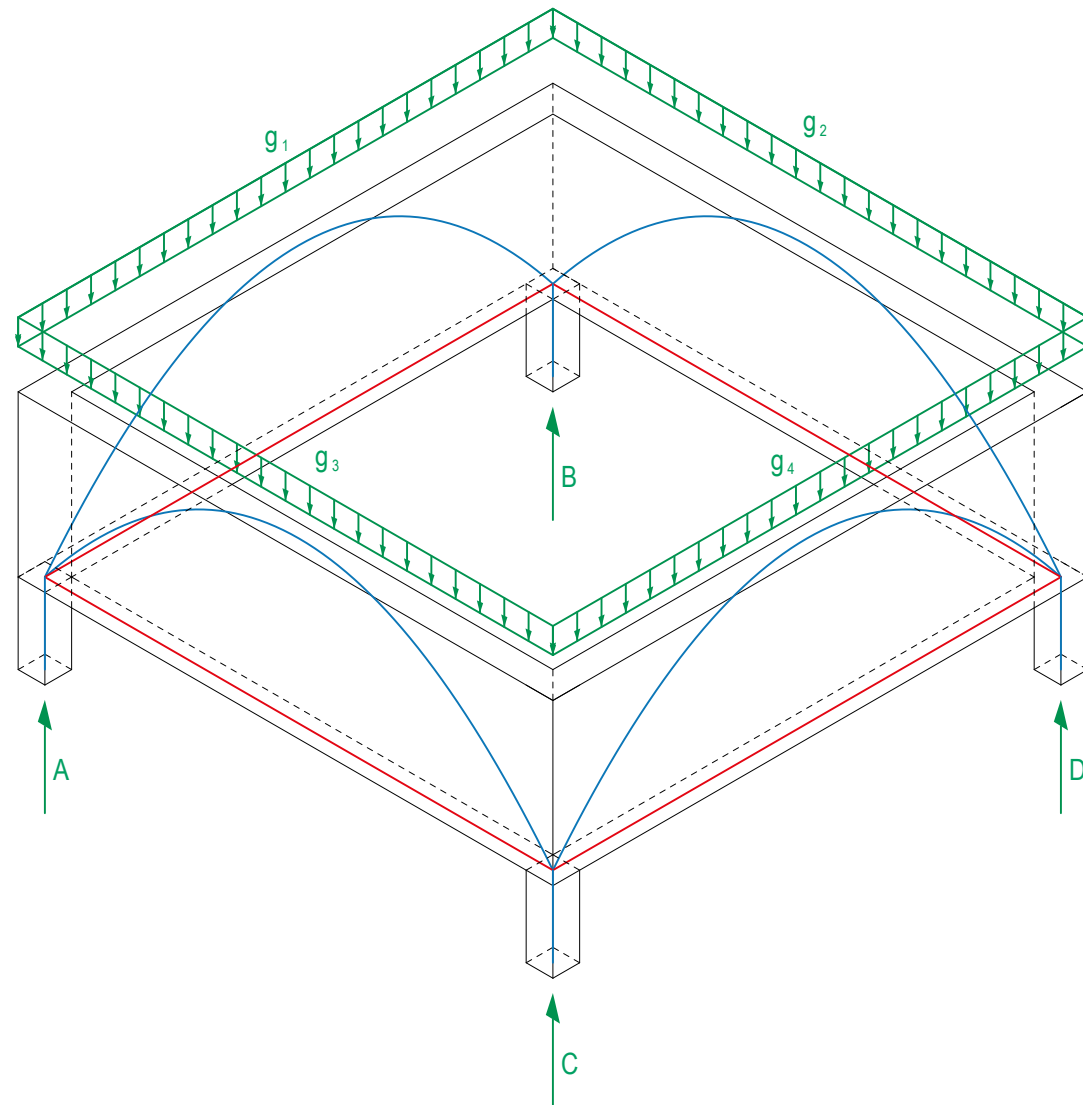
A	12.5 m²
$\bar{s}_d$	1.35 kN/m²
R	16.88 kN
g	2.25 kN/m

$$A = 2.5\text{m} \cdot 5\text{m} = 12.5 \text{ m}^2$$
$$R = \bar{s}_d \cdot A = 1.35 \text{ kN/m}^2 \cdot 12.5 \text{ m}^2 = 16.88 \text{ kN}$$
$$g = R / l = 16.88 \text{ kN} / 7.5 \text{ m} = 2.25 \text{ kN/m}$$

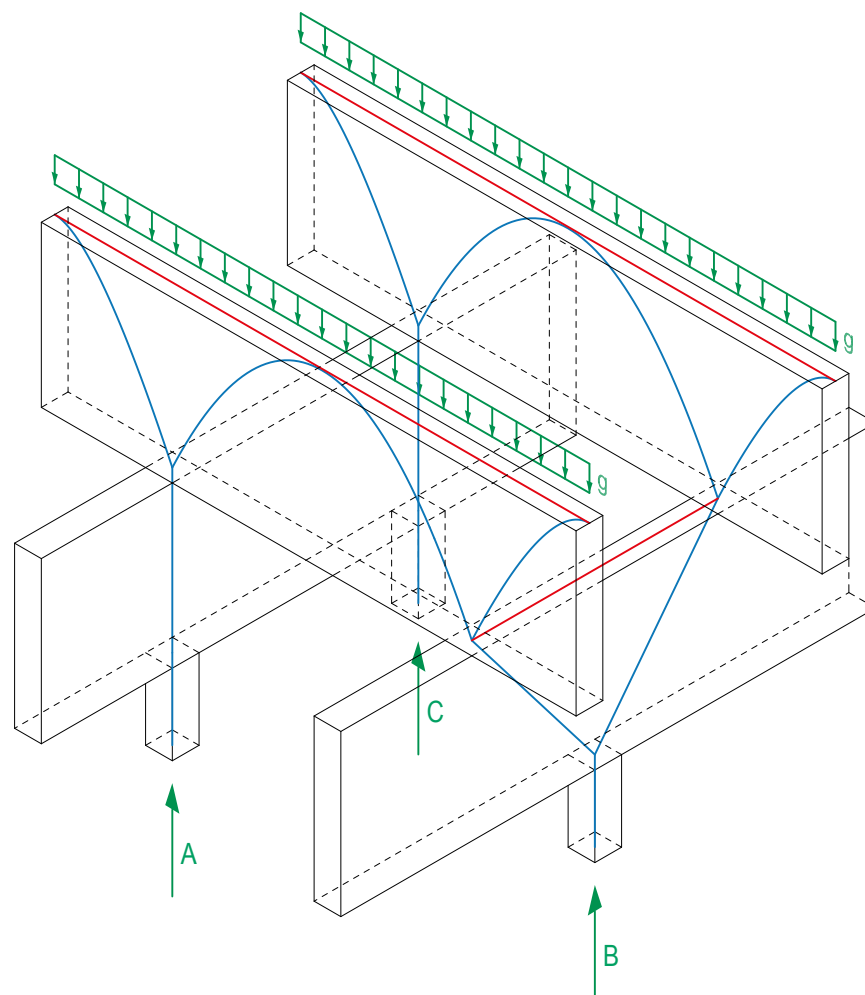
Task 3 Qualitative internal Force Flow

Draw a qualitative internal force flow in the axonometric drawing of the supporting structure. Use red for tension, blue for compression and green for the external forces.

a)



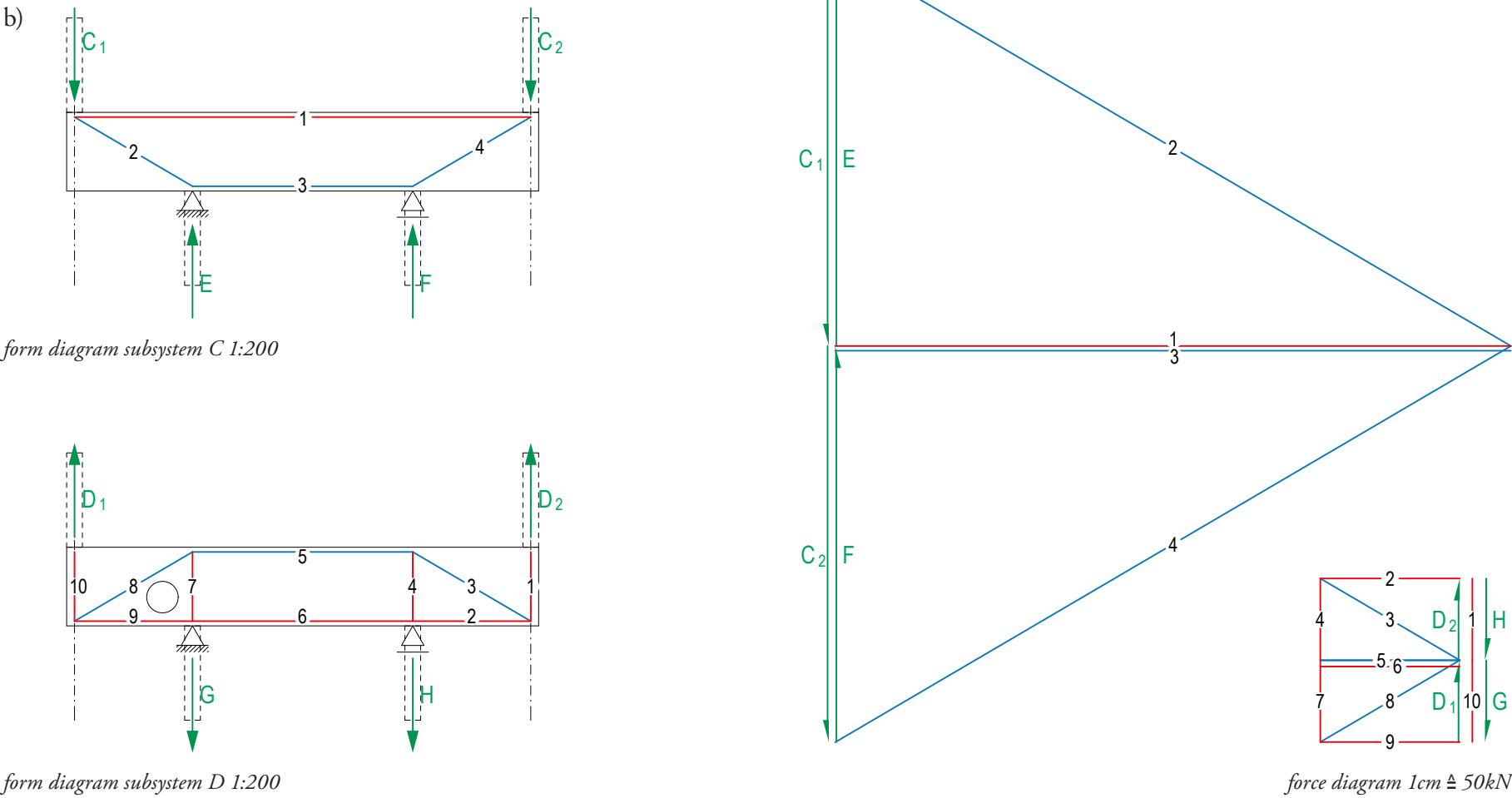
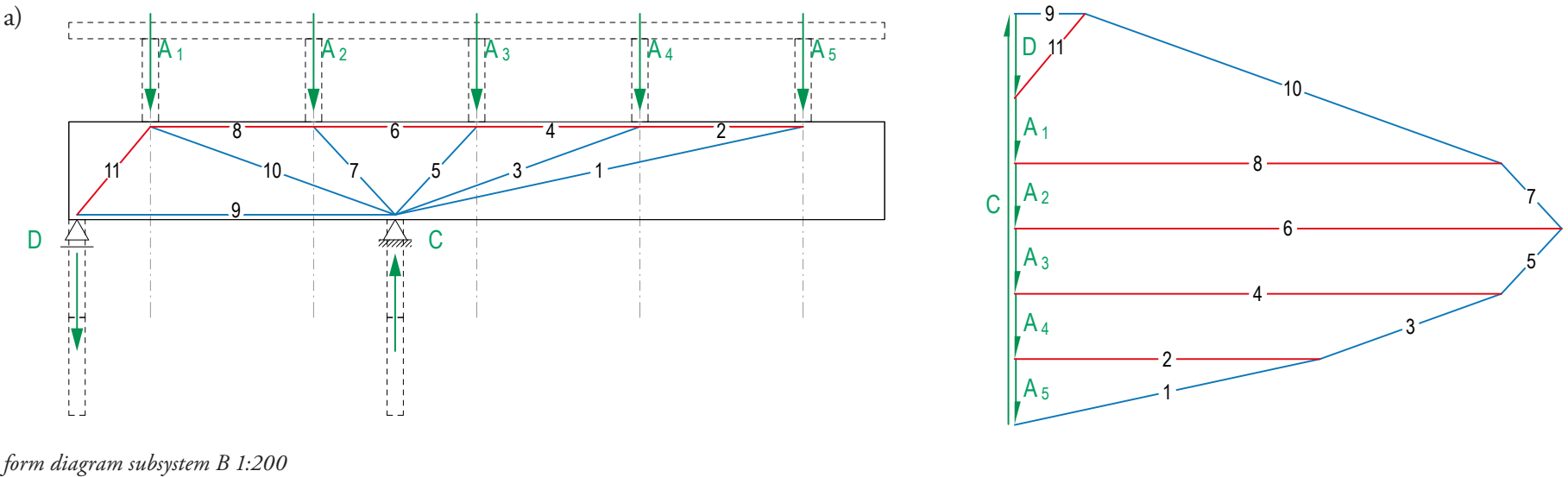
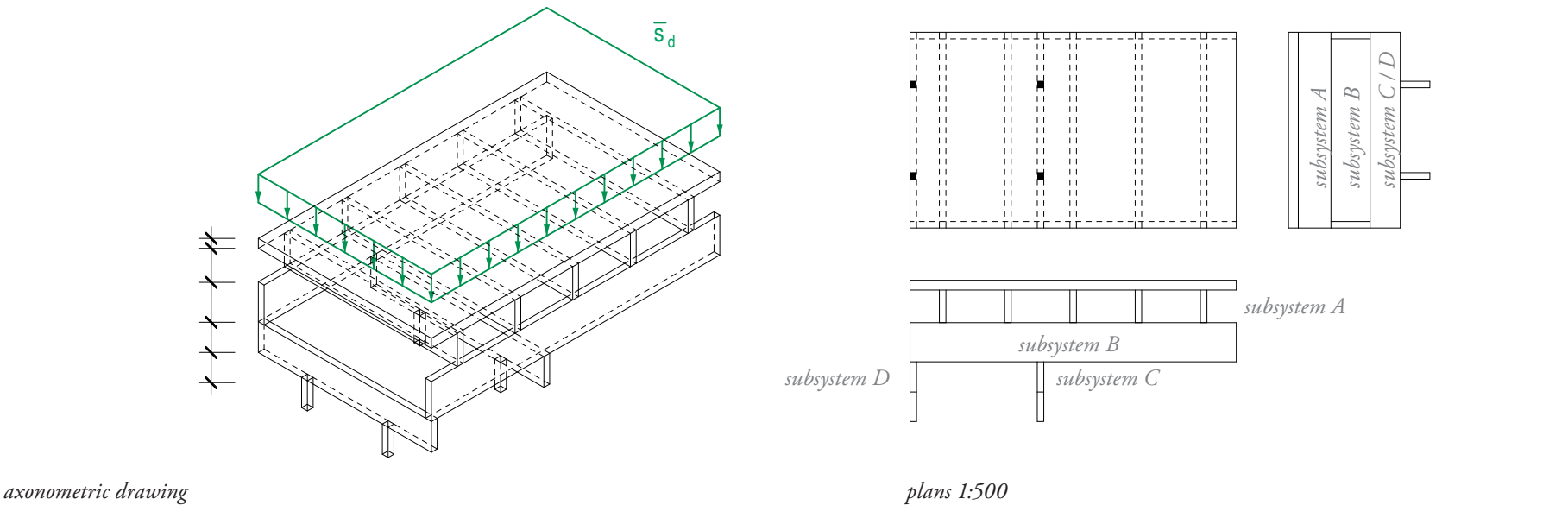
b)



Additional Transferring vertical Loads

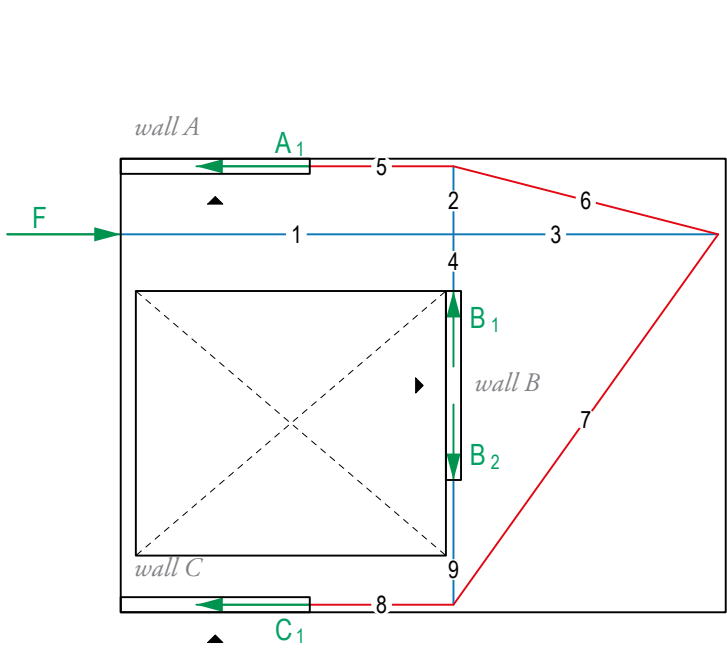
Task 1 The plate is loaded by the same dead area load of task 4.

- a) The reaction forces of the five beams in task 4a) (subsystem A) are further transferred to two other beams (Subsystem B). Note the different support condition of subsystem B. Find the inner force flow by the aid of the force diagram. Use red for tension, blue for compression and green for the reaction forces.
- b) Again the reaction forces found in a) are further transferred to subsystems C and D. Draw the external forces in the marked position. Find an internal force flow and draw the corresponding force diagram.

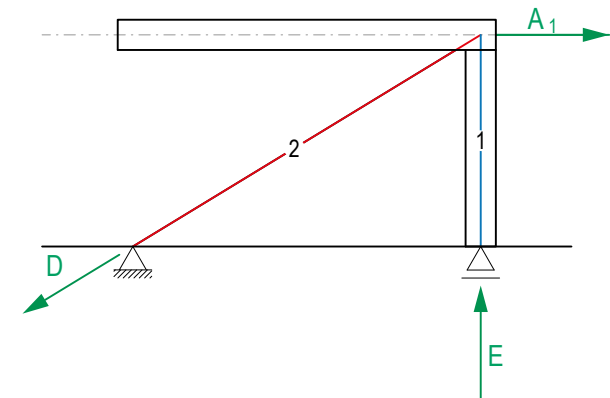
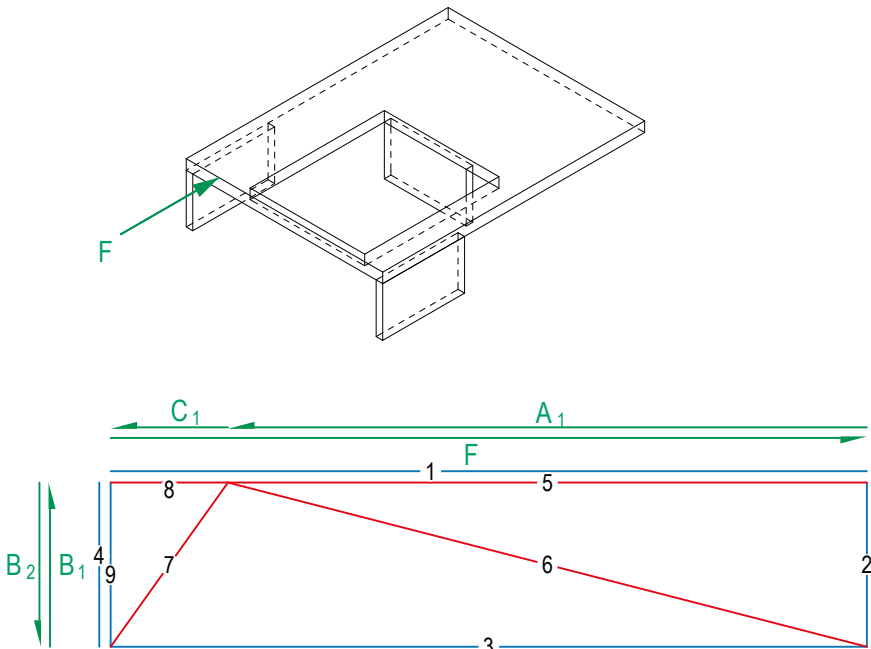


Additional Horizontal Forces (quantitatively)

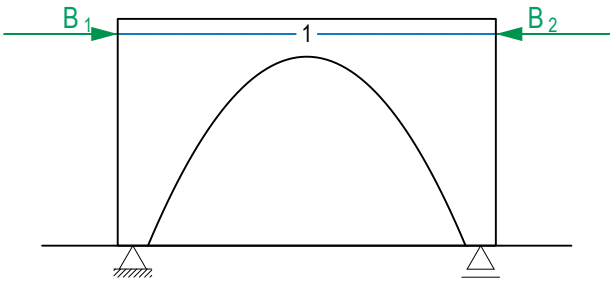
Task 1 Analyse the force flow within the ceiling due to an applied horizontal force. The walls are used for bracing, therefore find the internal force flow in the plate such as it can be redirected into the ground through the walls. Draw the corresponding force diagram if $F = 100\text{ kN}$.
Then find a possible internal force flow in the walls A, B and C. First draw the applied horizontal force for each wall into the corresponding form diagram. Use red for tension, blue for compression and green for the external forces.



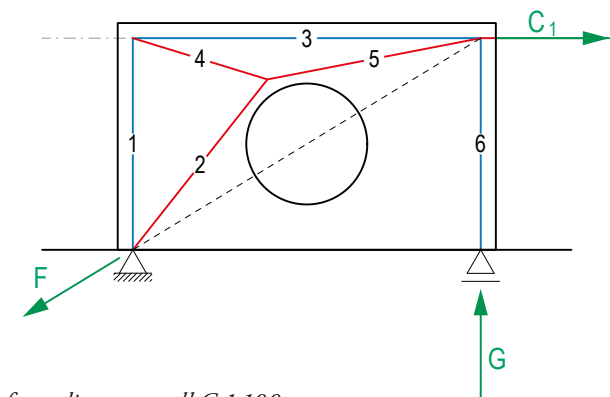
top view 1:200



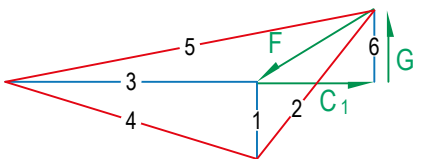
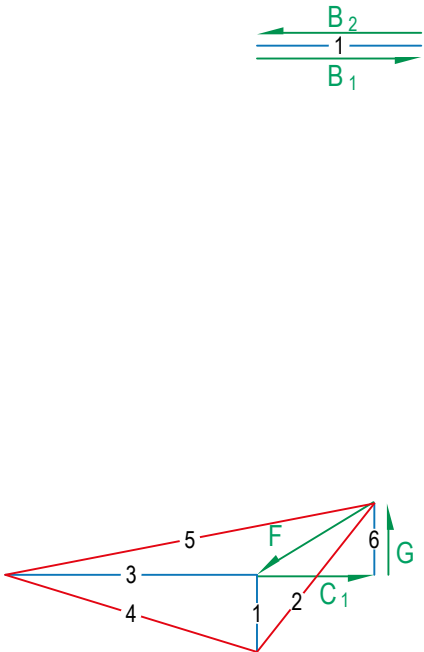
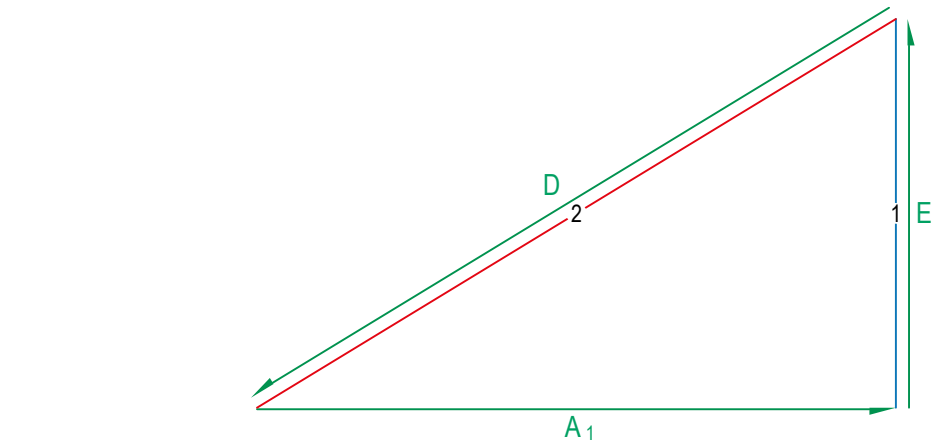
form diagram wall A 1:100



form diagram wall B 1:100



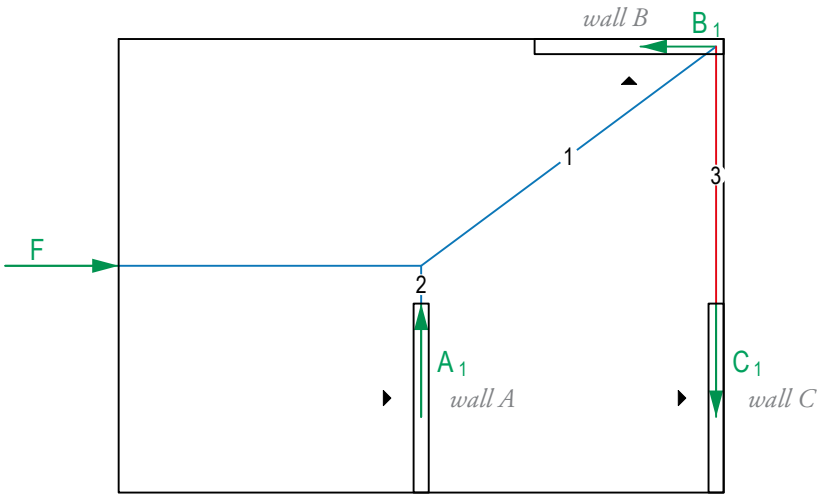
form diagram wall C 1:100



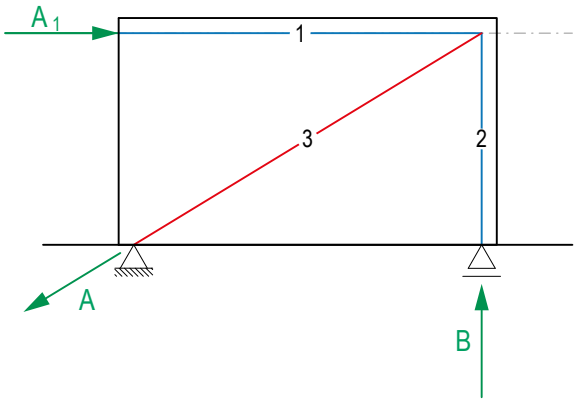
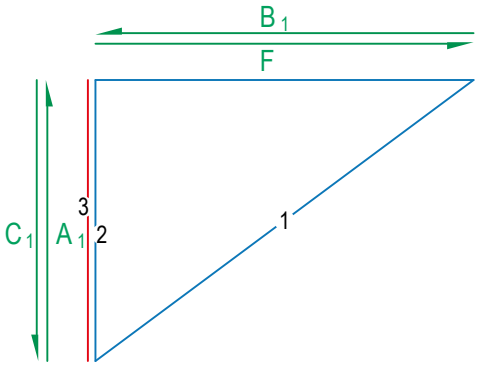
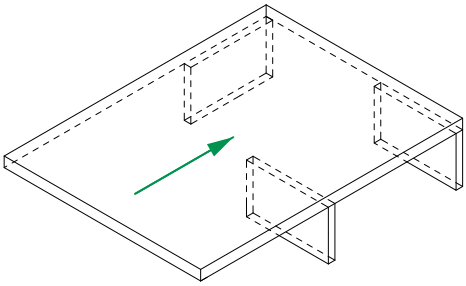
force diagrams 1cm ≙ 10kN

Additional Horizontal Forces (quantitatively)

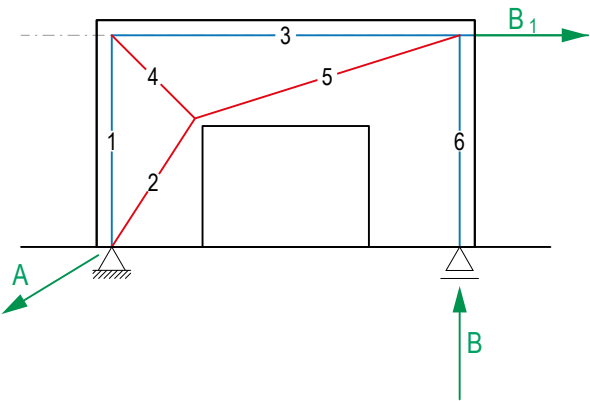
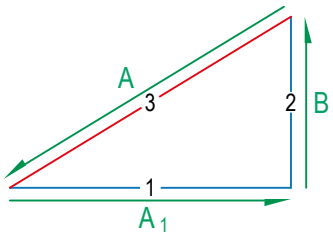
Task 3 Consider the force flow within the ceiling due to an applied horizontal force. The walls are used for bracing, therefore find the internal force flow in the plate such as it can be redirected into the ground through the walls. Draw the corresponding force diagram if $F = 100\text{ kN}$.
Then find a possible internal force flow in the walls A, B and C. First draw the applied horizontal force for each wall into the corresponding form diagram. Use red for tension, blue for compression and green for the external forces.



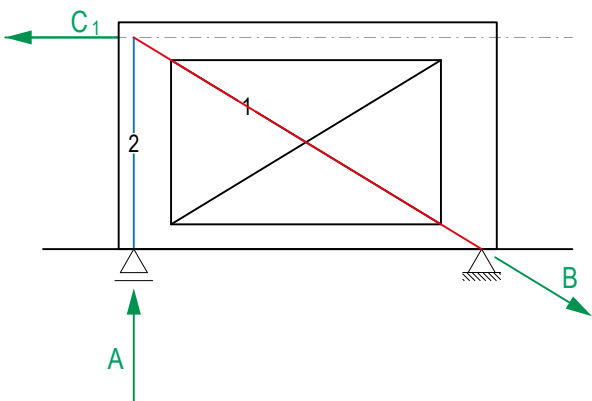
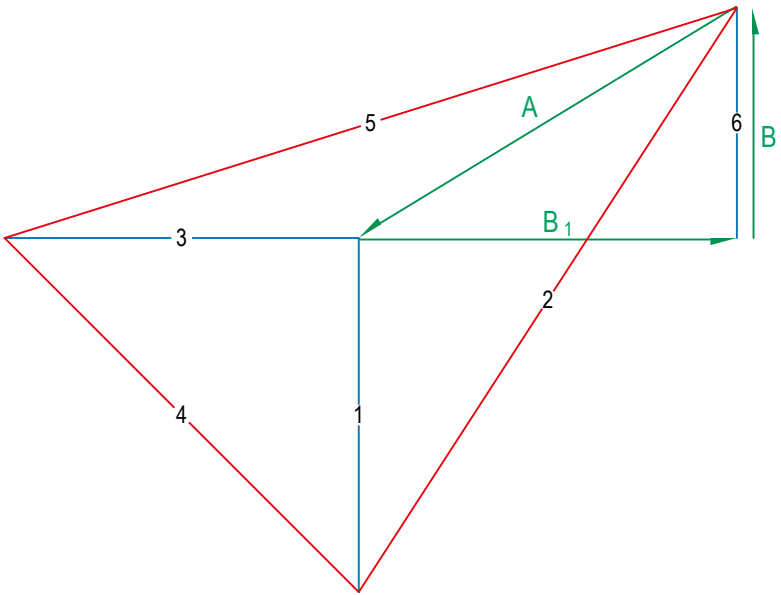
top view 1:200



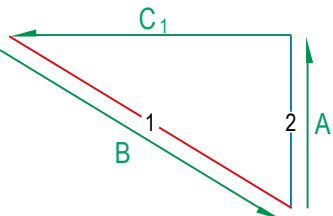
form diagram wall A 1:100



form diagram wall B 1:100



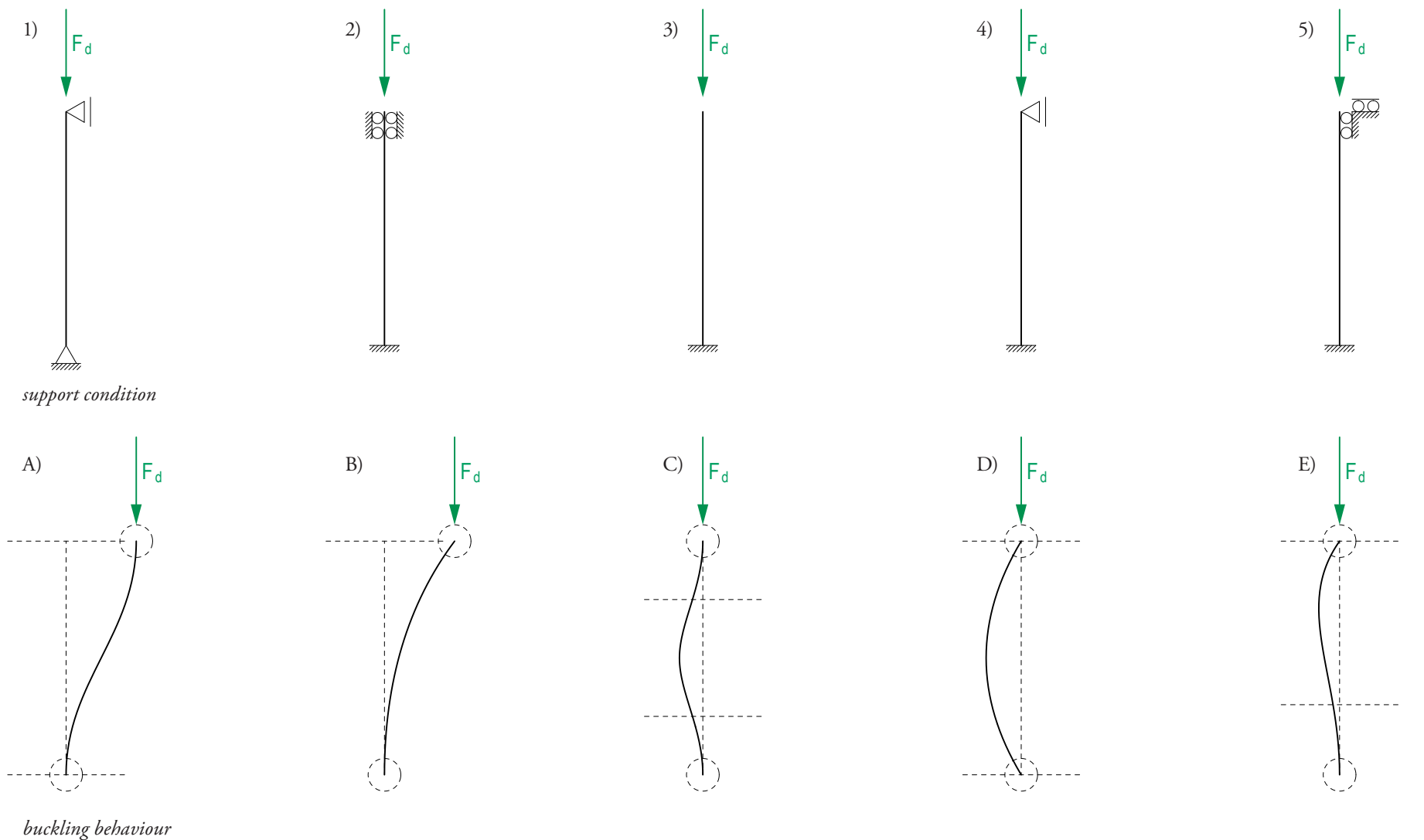
form diagram wall C 1:100



force diagrams 1cm ≙ 20kN

Task 1 Buckling behaviour

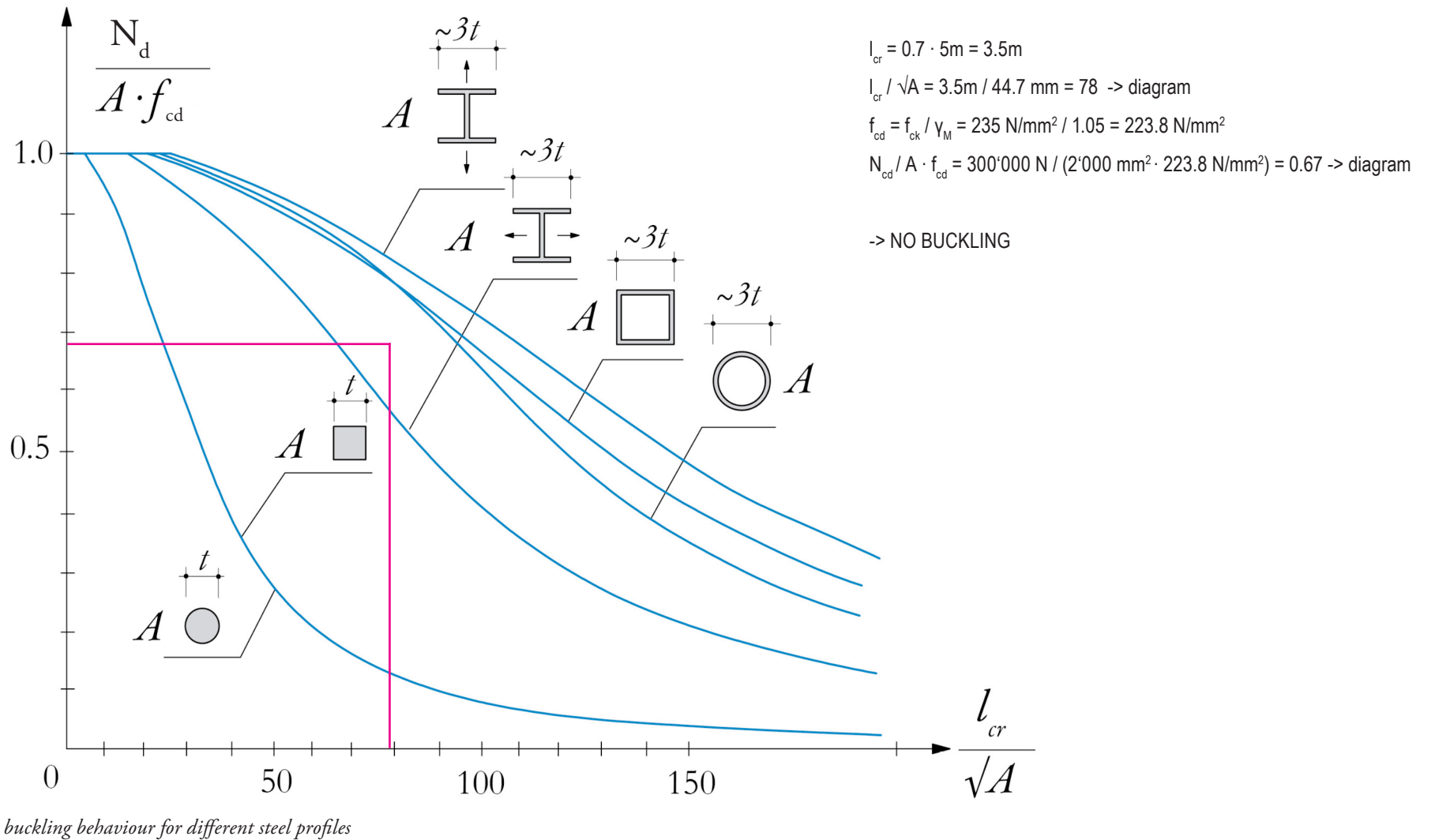
Shown below are columns with different supporting conditions and buckling behaviour. In the following table, assign to each support condition (1 - 5) the respective buckling behaviour (A - E) and the ratio of the critical and the actual length.



support condition	1	2	3	4	5
buckling behaviour	D	C	B	E	A
l_{cr} / l	1	0.5	2	0.7	1

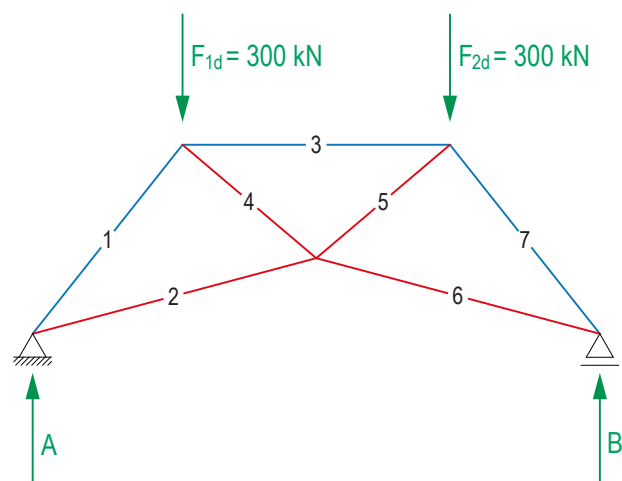
Task 2 Buckling test

We are looking at a square hollow profile out of steel S 235. The element has a length of 5 m, a cross section of $A = 2'000 \text{ mm}^2$ and is stressed by a compression force of $N_{cd} = 300 \text{ kN}$. The support condition corresponds to that of example 4) in task 1. Calculate whether buckling is occurring and draw the value in the diagram.

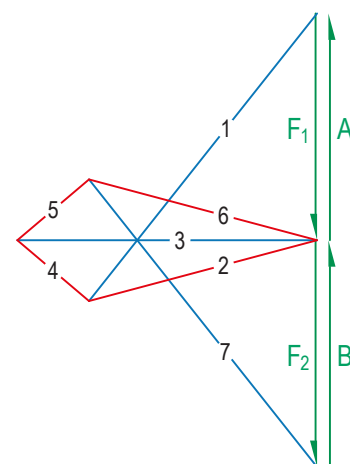


Task 3 Stability and dimensioning

a) Find the internal forces for the given structure with help of the force diagram. Use red for tension, blue for compression and green for the external forces.



form diagram 1:100



force diagram 1cm ≙ 100kN

b) The truss is built out of steel S 235 with circular solid profiles. Find the required diameter of the elements 1 and 2 (without considering buckling yet). Round the result to whole mm.

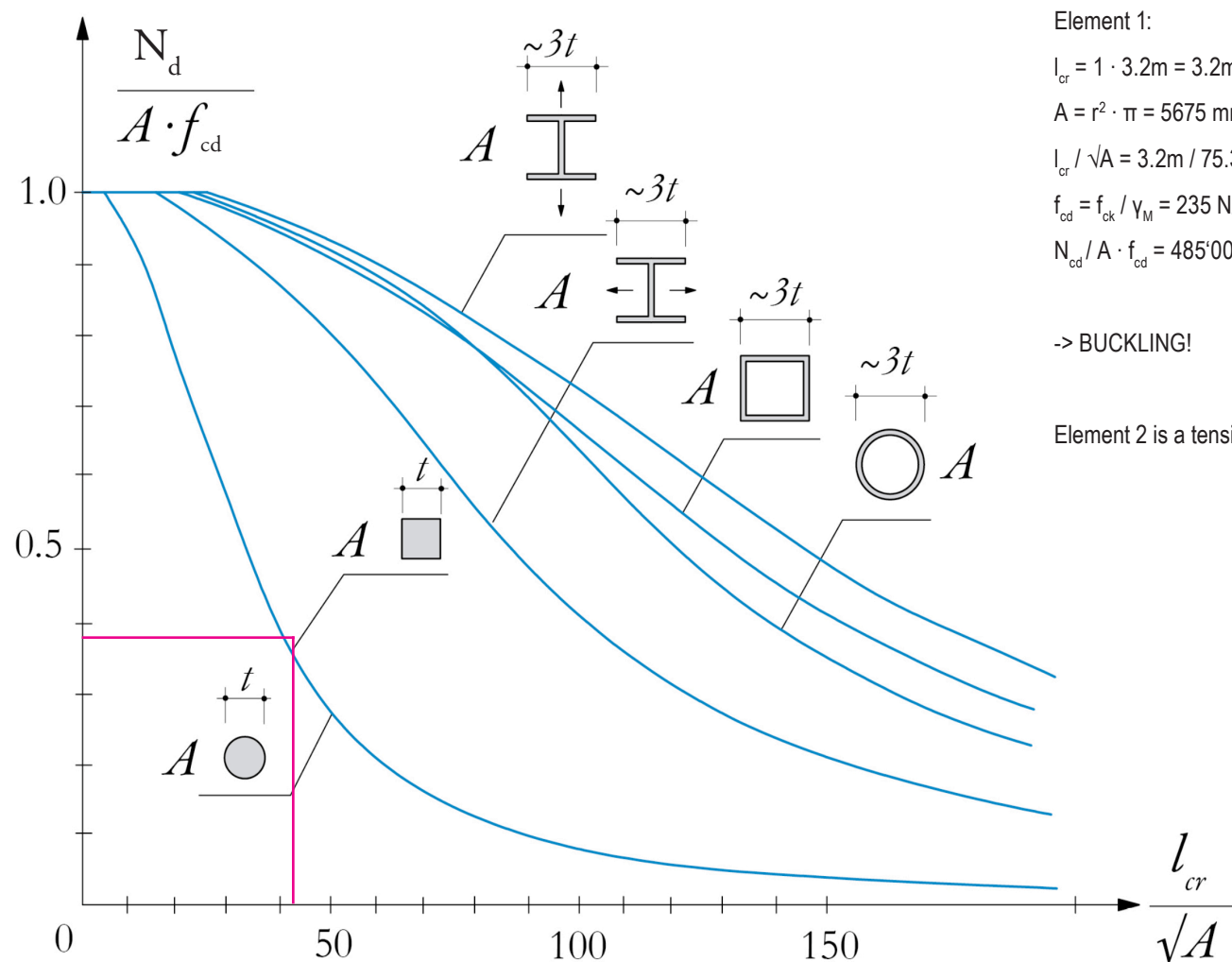
c) After considering the self-weight of the structure, the diameter for each element is 85 mm. In the given case the critical length is equal to the actual length of the elements. Use the buckling diagram to test the stability of elements 1 and 2 and draw the values into the diagram.

$$A_{\text{req}} = N_d / f_{\text{cd}} = 485 \text{ kN} / (235 \text{ N/mm}^2 / 1.05) = 2167 \text{ mm}^2$$

$$D_1 = 2 \cdot \sqrt{(A / \pi)} = 52.53 \text{ mm} \approx 53 \text{ mm}$$

$$A_{\text{req}} = N_d / f_{\text{td}} = 312 \text{ kN} / (235 \text{ N/mm}^2 / 1.05) = 1394 \text{ mm}^2$$

$$D_2 = 2 \cdot \sqrt{(A / \pi)} = 42.13 \text{ mm} \approx 43 \text{ mm}$$



buckling behaviour for different steel profiles

Element 1:

$$l_{\text{cr}} = 1 \cdot 3.2 \text{ m} = 3.2 \text{ m}$$

$$A = r^2 \cdot \pi = 5675 \text{ mm}^2$$

$$l_{\text{cr}} / \sqrt{A} = 3.2 \text{ m} / 75.3 \text{ mm} = 42.5 \rightarrow \text{diagram}$$

$$f_{\text{cd}} = f_{\text{ck}} / \gamma_M = 235 \text{ N/mm}^2 / 1.05 = 223.8 \text{ N/mm}^2$$

$$N_d / A \cdot f_{\text{cd}} = 485'000 \text{ N} / (5675 \text{ mm}^2 \cdot 223.8 \text{ N/mm}^2) = 0.38 \rightarrow \text{diagram}$$

-> BUCKLING!

Element 2 is a tensile element and therefore doesn't buckle.

d) How could buckling be prevented?

- use a hollow instead of a solid section profile
- choose a different material (e.g. S355 or S500)
- change the geometry (shorten the element)

Additional Transferring vertical loads

Task 3 In additional task 1, you have determined the maximum load of a column of the system illustrated below. Find the maximum roof load g based on the maximum load of the two columns.

- First draw a possible internal force flow in the lower beam as well as the corresponding force diagram. Find the maximum force that the two crossbeams can apply on the longitudinal beam. The magnitude of the reaction forces corresponds to the compression load found in additional task 1. Use red for tension, blue for compression and green for support forces.
- Then use the force diagram to find the internal force flow and the maximum line load g acting on the two crossbeams.

